

# TOPOLOGICAL AND LOGICAL ASPECTS OF THE PROBLEM OF HIDDEN VARIABLES IN QUANTUM MECHANICS

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**ABSTRACT.** We examine two proofs of the Bell–Kochen–Specker theorem, which rules out non-contextual hidden variables in quantum mechanics. Bell’s proof combines Gleason’s theorem with elementary topology and is infinitary; Kochen and Specker’s proof exhibits a finite configuration of rays and is constructive. The two arguments were found independently in the mid-1960s. We show that the compactness theorem of propositional logic links them: compactness turns Bell’s argument into the existence of a finite configuration of the kind that Kochen and Specker built by hand. We also show that quantum measures have no counterpart of Dirac measures and, unlike classical measures, do not form a lattice. A timeline traces the history of the problem of hidden variables.

## 0. PRELIMINARIES (ALL THE LINEAR ALGEBRA AND PHYSICS YOU WILL NEED TO READ THESE NOTES)

We assume that the reader is familiar with Hilbert spaces, with eigenvalues and eigenvectors of linear operators, and with orthogonal projections. Everything else we need, from both linear algebra and physics, is recalled in this short section.

Unless stated otherwise (as in Remark 1.4 and Section 6), the symbol  $H$  denotes a complex Hilbert space with  $3 \leq \dim H \leq \infty$ , where  $\infty$  stands for  $\aleph_0$ : unless stated otherwise, Hilbert spaces are separable. (Separability is not a technicality; Remark 1.5 below explains what large cardinals have to do with it.) Everything below holds verbatim for real and quaternionic Hilbert spaces as well.

We use the physicists’ notation for vectors and functionals, which we now explain, since this notation is not standard in mathematics. The inner product  $\langle \xi | \eta \rangle$  of  $\xi, \eta \in H$  is linear in the second variable and conjugate-linear in the first. A vector  $\eta$  is also written  $|\eta\rangle$  (a *ket*), and  $\langle \xi |$  (a *bra*) denotes the linear functional  $\eta \mapsto \langle \xi | \eta \rangle$ . Juxtaposition is composition: for a bounded operator  $A$ , the number  $\langle \xi | A | \eta \rangle$  means  $\langle \xi | A \eta \rangle$ , and  $|\xi\rangle \langle \xi |$  is the operator  $\eta \mapsto \langle \xi | \eta \rangle \xi$ . We write  $S(H)$  for the unit sphere of  $H$ .

Let  $L(H)$  be the lattice of closed subspaces of  $H$  — an *orthomodular lattice* in the usual sense — which we identify with the corresponding orthogonal projections. Recall that the orthogonal projection  $P$  onto a closed subspace  $V$  is defined by writing each  $\eta \in H$  uniquely as a sum  $\eta = \eta_V + \eta_\perp$  with  $\eta_V \in V$  and  $\eta_\perp \in V^\perp$ , and putting  $P\eta = \eta_V$ . Two properties follow at once. First, the projection  $P$  satisfies  $P^2 = P$ , because  $P$  fixes every vector of  $V$ . Second, the projection  $P$  satisfies  $P^* = P$ , because for  $\xi, \eta \in H$  we have

$$\langle \xi | P \eta \rangle = \langle \xi_V | \eta_V \rangle = \langle P \xi | \eta \rangle,$$

since the cross terms  $\langle \xi_\perp | \eta_V \rangle$  and  $\langle \xi_V | \eta_\perp \rangle$  vanish. Conversely, a bounded operator  $P$  with  $P^2 = P = P^*$  is the orthogonal projection onto the range of  $P$ , which is closed. Thus, the elements of  $L(H)$  are exactly the bounded operators  $P$  with  $P^2 = P = P^*$ , and from these two identities

$$\langle \eta | P | \eta \rangle = \langle \eta | P^* P | \eta \rangle = \langle P \eta | P \eta \rangle = \|P \eta\|^2 \quad (\eta \in H). \quad (1)$$

The lattice  $L(H)$  carries a partial order  $P \leq Q$  (inclusion of ranges), a meet  $P \wedge Q$  (intersection of ranges), a join  $P \vee Q$  (closed span of the ranges), an orthocomplement  $P \mapsto P^\perp = I - P$ , and extremal elements 0 and  $I = H$ . Two projections  $P$  and  $Q$  are *orthogonal*, written  $P \perp Q$ , when their ranges are orthogonal, equivalently  $P \leq Q^\perp$ . A *basis* of  $L(H)$  is the set of rank-one projections onto the vectors of some orthonormal basis of  $H$ .

For  $\psi \in S(H)$ , the operator  $P_\psi = |\psi\rangle\langle\psi|$  is the orthogonal projection onto the one-dimensional subspace  $\mathbb{C}\psi$ : the projection  $P_\psi$  fixes  $\psi$  and kills every vector orthogonal to  $\psi$ , and  $P_\psi^2 = P_\psi = P_\psi^*$ . The rank-one projections are exactly the minimal nonzero elements, the *atoms*, of the lattice  $L(H)$ .

We now turn to the physical meaning of these objects. In quantum mechanics, a projection  $P \in L(H)$  represents a yes/no question about a physical system, that is, an experiment with two possible outcomes, and  $P^\perp = I - P$  represents the same question with the answers exchanged. A state of the system assigns to each projection  $P$  the probability  $\mu(P)$  of obtaining the answer “yes” to  $P$ . The question  $I$  is always answered “yes”, so  $\mu(I) = 1$ . If  $P_0, P_1, \dots$  are pairwise orthogonal, the corresponding questions are mutually exclusive and can be tested together, and the probability of “yes” to their join  $\bigvee_i P_i$  is the sum of the probabilities  $\mu(P_i)$ . These requirements are the axioms of a quantum measure (see Definition 1.1 below). Which assignments  $\mu$  actually occur in physics is decided by the Born rule, which leads to the notion of density operator.



**Remark 0.1 (The Born rule).** The simplest states of a physical system, the *pure states*, are described by unit vectors  $\psi \in S(H)$ . The *Born rule* says that in the state  $\psi$  the probability of the answer “yes” to the question  $P$  is

$$\mu(P) = \|P\psi\|^2 = \langle\psi|P|\psi\rangle$$

(the second equality is (1)). An important special case is the question “is the system in the state  $\xi$ ?” for another unit vector  $\xi \in S(H)$ . This question is represented by the rank-one projection  $P_\xi = |\xi\rangle\langle\xi|$  onto the line  $\mathbb{C}\xi$ , which maps a vector  $\eta$  to  $\langle\xi|\eta\rangle\xi$ . Since  $P_\xi\psi = \langle\xi|\psi\rangle\xi$  and  $\|\xi\| = 1$ , the Born rule gives

$$\mu(P_\xi) = \|P_\xi\psi\|^2 = |\langle\xi|\psi\rangle|^2,$$

the familiar probability that a system in the state  $\psi$  is found in the state  $\xi$ .

A pure state represents complete knowledge of the system: no further information about how the system was prepared could sharpen the probabilities  $\|P\psi\|^2$ . Even so, these probabilities are in general strictly between 0 and 1. The randomness they describe is therefore not due to any lack of information on our part; according to quantum mechanics this randomness is intrinsic to the system itself, and the theorems of these notes are precisely about whether this randomness could be explained away as ignorance of some hidden variables.

In practice, however, even this complete knowledge is rarely available. Real physical sources do not produce a single pure state: the light of an ordinary lamp, for instance, consists of photons whose polarizations vary at random from one photon to the next, so no single vector describes the beam. Our information about how a system was prepared is usually incomplete, and the imperfections of the preparing apparatus mean that we know at best which pure states may have been produced, and with what probabilities. This introduces a second kind of randomness, of an entirely different nature from the first: this randomness is classical, of the same kind as in a game of cards, and reflects only our ignorance, that is, the

imperfection of our information about the system, not any property of the system itself. A state that combines the two kinds of randomness is called a *mixed state*, or a *mixture*.<sup>1</sup>

Accordingly, suppose one knows only that the system has been prepared in the state  $\psi_k$  with probability  $\lambda_k$ , for a family of states  $\psi_k \in S(H)$  and weights  $\lambda_k \geq 0$  with  $\sum_k \lambda_k = 1$ . Here the weights  $\lambda_k$  express our ignorance of the preparation, while each  $\langle \psi_k | P | \psi_k \rangle$  is a quantum probability given by the Born rule. Combining the two, the probability of “yes” to  $P$  is then the weighted average (or expected value)

$$\sum_k \lambda_k \langle \psi_k | P | \psi_k \rangle. \quad (2)$$

We now show that this expression depends on the preparation only through the operator

$$\rho = \sum_k \lambda_k |\psi_k\rangle\langle\psi_k| = \sum_k \lambda_k P_{\psi_k},$$

and we look for a single formula that computes (2) from  $\rho$  and  $P$ . The trace provides this formula, as we now explain.

When  $H$  is finite-dimensional, the *trace* of a linear operator  $T$  on  $H$  is

$$\text{Tr}(T) = \sum_i \langle e_i | T e_i \rangle,$$

where  $(e_i)$  is an orthonormal basis of  $H$ . Thus  $\text{Tr}(T)$  is the sum of the diagonal entries of the matrix of  $T$  in the basis  $(e_i)$ , a sum that does not depend on the choice of basis. When  $H$  is infinite-dimensional, the same formula defines the trace of the *trace-class* operators, for which the sum converges absolutely and is again independent of the basis. We need only two facts about trace-class operators: every operator of finite rank, in particular every rank-one projection  $P_\psi$ , is trace-class; and if  $T$  is trace-class and  $A$  is bounded, then  $TA$  and  $AT$  are trace-class. For the definition and these facts, see [RS80, Section VI.6]. The reader who prefers may assume that  $H$  is finite-dimensional throughout this section; the infinite-dimensional case differs only in these points of analysis.

The following observation will play an important role in these notes.

**Observation 0.2.** *Let  $A$  be a bounded operator on  $H$  and let  $\psi \in S(H)$  be given. Then*

$$\text{Tr}(AP_\psi) = \text{Tr}(P_\psi A) = \langle \psi | A | \psi \rangle.$$

*In particular, we have  $\text{Tr}(P_\psi P) = \langle \psi | P | \psi \rangle = \|P\psi\|^2$  for every  $P \in L(H)$ , and  $\text{Tr}(P_\xi P_\psi) = |\langle \xi | \psi \rangle|^2$  for  $\xi \in S(H)$ .*

*Proof.* Compute the trace in an orthonormal basis  $(e_i)$  of  $H$  with  $e_0 = \psi$ . Then  $P_\psi e_0 = \psi$ , and  $P_\psi e_i = 0$  for  $i \geq 1$  because  $e_i$  is orthogonal to  $\psi$ . So only the first term survives:

$$\text{Tr}(AP_\psi) = \sum_i \langle e_i | AP_\psi e_i \rangle = \langle \psi | A \psi \rangle = \langle \psi | A | \psi \rangle.$$

Similarly, since  $P_\psi A e_i = \langle \psi | A e_i \rangle \psi$  and  $\langle e_i | \psi \rangle = 0$  for  $i \geq 1$ , we have

$$\text{Tr}(P_\psi A) = \sum_i \langle e_i | P_\psi A e_i \rangle = \sum_i \langle e_i | \psi \rangle \langle \psi | A e_i \rangle = \langle \psi | A \psi \rangle.$$

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<sup>1</sup>The same mathematical description arises in a situation where ignorance is not the explanation: a system  $A$  that has interacted with another system  $B$ , and is therefore entangled with  $B$ , has in general no pure state of  $A$  alone, even when the combined system is in a pure state, and the statistics of  $A$  are again those of a mixture.

If  $P \in L(H)$ , then by (1)  $\langle \psi | P \psi \rangle = \langle P \psi | P \psi \rangle = \|P \psi\|^2$ . Finally  $P_\psi \xi = \langle \psi | \xi \rangle \psi$ , so taking  $A = P_\xi$  gives  $\text{Tr}(P_\xi P_\psi) = \|P_\psi \xi\|^2 = |\langle \xi | \psi \rangle|^2$ . ☺

Returning to the Born rule: by Observation 0.2 and linearity of the trace, the probability (2) is

$$\sum_k \lambda_k \langle \psi_k | P | \psi_k \rangle = \sum_k \lambda_k \text{Tr}(P_{\psi_k} P) = \text{Tr}(\rho P).$$

So the Born rule, for pure and mixed preparations alike, takes the single form  $P \mapsto \text{Tr}(\rho P)$ . The operator  $\rho$  has three properties that do not depend on how  $\rho$  was obtained. First, the operator  $\rho$  is positive, since  $\langle \xi | \rho | \xi \rangle = \sum_k \lambda_k |\langle \psi_k | \xi \rangle|^2 \geq 0$ . Second, the operator  $\rho$  is trace-class. Third, the trace of  $\rho$  is 1; this property is nothing but the requirement that the weights  $\lambda_k$  add up to 1: each  $P_{\psi_k}$  has trace  $\text{Tr}(P_{\psi_k}) = \langle \psi_k | \psi_k \rangle = 1$  (take  $A = I$  in Observation 0.2), so

$$\text{Tr}(\rho) = \sum_k \lambda_k \text{Tr}(P_{\psi_k}) = \sum_k \lambda_k = 1.$$

Equivalently, the number  $\text{Tr}(\rho) = \text{Tr}(\rho I)$  is the probability of the answer “yes” to the question  $I$ , which is always answered “yes”. These are the properties singled out in the following definition.

**Definition 0.3 (Density operator).** A *density operator* (or *density matrix*) on  $H$  is a trace-class operator  $\rho$  that is positive, i.e.  $\langle \xi | \rho | \xi \rangle \geq 0$  for all  $\xi \in H$ , and has  $\text{Tr}(\rho) = 1$ .

Conversely, every density operator arises from a preparation as in Remark 0.1. Suppose first that  $H$  is finite-dimensional. A positive operator  $\rho$  on a complex Hilbert space is self-adjoint, so  $\rho$  has an orthonormal basis of eigenvectors  $\psi_k$ ; the corresponding eigenvalues  $\lambda_k = \langle \psi_k | \rho | \psi_k \rangle$  are nonnegative because  $\rho$  is positive, and they add up to  $\text{Tr}(\rho) = 1$ . Hence  $\rho$  can be written as

$$\rho = \sum_k \lambda_k |\psi_k\rangle\langle\psi_k|, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1, \quad (3)$$

where  $\psi_k$  has eigenvalue  $\lambda_k$ . When  $H$  is infinite-dimensional, the same decomposition holds with a countable orthonormal family  $(\psi_k)$  of eigenvectors, by the spectral theorem for compact self-adjoint operators [RS80, Section VI.5] (density operators, being trace-class, are compact). So density operators are exactly the probabilistic mixtures of the *pure states*  $P_\psi$  with  $\psi \in S(H)$ .

Finally, taking  $A = \rho$  in Observation 0.2, the probability of finding a system in the state  $\rho$  in the pure state  $\psi$  is

$$\text{Tr}(\rho P_\psi) = \langle \psi | \rho | \psi \rangle.$$

If  $\psi$  is an eigenvector of  $\rho$  with eigenvalue  $\lambda$ , then  $\langle \psi | \rho | \psi \rangle = \lambda \|\psi\|^2 = \lambda$ ; in general, the number  $\langle \psi | \rho | \psi \rangle$  is a weighted average of the eigenvalues of  $\rho$ , namely  $\sum_k \lambda_k |\langle \psi_k | \psi \rangle|^2$ .

**That is all, dear reader.** You now have all the quantum mechanics and linear algebra needed for the rest of these notes.

## 1. QUANTUM MEASURES AND GLEASON’S THEOREM

In this section we define quantum measures, state the main theorem of these notes, and recall Gleason’s Theorem, which describes all quantum measures when  $\dim H \geq 3$ .

**Definition 1.1 (Quantum measure).** A *quantum measure* (or *state*<sup>2</sup>) on  $L(H)$  is a function  $\mu : L(H) \rightarrow [0, 1]$  with  $\mu(I) = 1$  that is countably additive on orthogonal families: for pairwise orthogonal  $\{P_i\}$ , we have

$$\mu\left(\bigvee_i P_i\right) = \sum_i \mu(P_i).$$

A *two-valued quantum measure* is a quantum measure  $v : L(H) \rightarrow \{0, 1\}$ .

**What a two-valued quantum measure represents.** Definition 1.1 is purely mathematical: the definition concerns functions on a lattice and mentions nothing physical. Read physically, however, a two-valued quantum measure expresses a precise hypothesis about nature. Recall from Section 0 that a projection  $P$  represents a yes/no question about a physical system. A two-valued quantum measure  $v$  answers every such question in advance: the answer to  $P$  is “yes” if  $v(P) = 1$  and “no” if  $v(P) = 0$ , and this answer does not depend on which other compatible questions are asked together with  $P$ . Moreover, the answers to the questions of each orthonormal basis are consistent: exactly one of them is “yes” (see Lemma 2.1 below). The hypothesis that every yes/no question has such a predetermined answer is what physicists call a *non-contextual hidden-variable assignment*, and a two-valued quantum measure is also called a *valuation*. If valuations existed, the randomness of quantum mechanics could be explained as ignorance: a state would be an average of valuations, just as a classical probability measure is an average of Dirac measures, and quantum probabilities would merely record our uncertainty about which valuation describes the system. The main theorem of these notes rules out this explanation.

**Theorem 1.2 (Main Theorem, Bell–Kochen–Specker).** *If  $3 \leq \dim H \leq \infty$ , then no two-valued quantum measure exists on  $L(H)$ .*

We give two independent proofs of Theorem 1.2: a topological one, due to Bell (see Section 3 below; Bell’s proof is the simpler of the two) and a combinatorial one, due to Kochen and Specker (see Section 5 below). Bell’s proof relies on Gleason’s Theorem, which describes all quantum measures in terms of density operators (see Definition 0.3). For every density operator  $\rho$ , the map  $P \mapsto \text{Tr}(\rho P)$  is a quantum measure: this map takes values in  $[0, 1]$  (by (3) and Observation 0.2, we have  $\text{Tr}(\rho P) = \sum_k \lambda_k \|P\psi_k\|^2$ ), takes the value  $\text{Tr}(\rho) = 1$  at  $I$ , and is additive on finite orthogonal families, because the join of finitely many pairwise orthogonal projections is their sum and the trace is linear. In finite dimension every orthogonal family of nonzero projections is finite, so finite additivity is all that is needed. In infinite dimension, countable additivity follows in the same way from the continuity of  $A \mapsto \text{Tr}(\rho A)$  on bounded increasing sequences of projections [RS80, Section VI.6]. By Remark 0.1, these are exactly the quantum measures given by the Born rule. Gleason’s Theorem (see Theorem 1.3 below) says that, conversely, every quantum measure arises in this way: in dimension at least 3, the Born rule is forced by the lattice structure of  $L(H)$  alone.

We can now state Gleason’s Theorem.

**Theorem 1.3 (Gleason’s Theorem [Gle57]).** *If  $3 \leq \dim H \leq \infty$ , then every quantum measure  $\mu$  on  $L(H)$  has the form  $\mu(P) = \text{Tr}(\rho P)$  for a unique density operator  $\rho$  (see Definition 0.3).*

<sup>2</sup>Gleason called this object a *measure* [Gle57]; the quantum-logic tradition (Mackey [Mac63], Piron [Pir64]) relabeled this object a *state*, and much of the older literature still uses that name; later authors (e.g. [Ham03]) reverted to *quantum measure*, the term used throughout these notes.

We take Gleason's Theorem as given, since every known proof (Gleason's original argument, or any later streamlining) is a substantial piece of real analysis. The following remark, which comes from Gleason's paper [Gle57, p. 886], shows that the lower bound  $\dim H \geq 3$  cannot be weakened to  $\dim H \geq 2$ . Remark 1.4 is not needed for our proofs and can be skipped without loss of continuity.



**Remark 1.4.** We show that the lower bound  $\dim H \geq 3$  is optimal. On  $H = \mathbb{R}^2$ , parametrize rays by angle  $\theta \in [0, \pi)$ . Every orthonormal basis is a pair  $\{\theta, \theta + \pi/2 \pmod{\pi}\}$ . Define a function  $v : [0, \pi) \rightarrow \{0, 1\}$  by

$$v(\theta) = \begin{cases} 1, & \text{if } \theta \in [0, \pi/2), \\ 0, & \text{if } \theta \in [\pi/2, \pi). \end{cases}$$

In every such pair, exactly one of the two angles lands in  $[0, \pi/2)$ , so  $v$  satisfies additivity on every basis; in other words, the function  $v$  is a genuine quantum measure in the sense of Definition 1.1, two-valued, on  $L(\mathbb{R}^2)$  (with  $v(0) = 0$  and  $v(I) = 1$ ). The function  $v$  jumps at  $\theta = \pi/2$ , whereas  $\theta \mapsto \langle \theta | \rho | \theta \rangle$  is continuous for every density operator  $\rho$ ; so  $v$  is not of the form  $\text{Tr}(\rho \cdot)$ , and the conclusion of Gleason's Theorem fails when  $\dim H = 2$ . The same construction works for  $H = \mathbb{C}^2$ : in each pair of orthogonal rays, assign the value 1 to one ray and 0 to the other, and set  $v(0) = 0$  and  $v(I) = 1$ .

The upper bound on the dimension is just as sharp, but for a reason that has nothing to do with geometry: in nonseparable Hilbert spaces, Gleason's Theorem depends on large cardinals.



**Remark 1.5 (Gleason's Theorem and real-valued measurable cardinals).** Let  $H$  be a Hilbert space with an orthonormal basis  $\{e_i\}_{i \in I}$ , and suppose that  $|I|$  is at least a real-valued measurable cardinal, so that there is a countably additive probability measure  $\nu$  on the power set of  $I$  that vanishes on singletons. Put

$$\mu(P) = \int_I \langle e_i | P | e_i \rangle d\nu(i) \quad (P \in L(H)),$$

which makes sense because  $\nu$  measures every subset of  $I$ . We have  $\mu(I) = 1$ , and  $\mu$  is countably additive on orthogonal families: if  $P = \bigvee_k P_k$  with the  $P_k$  pairwise orthogonal, then  $\langle e_i | P | e_i \rangle = \sum_k \langle e_i | P_k | e_i \rangle$  for every  $i$ , and the monotone convergence theorem lets us integrate term by term. However,  $\mu(P_{e_i}) = \nu(\{i\}) = 0$  for every  $i$ , whereas a measure  $\text{Tr}(\rho \cdot)$  satisfies  $\sum_{i \in I} \text{Tr}(\rho P_{e_i}) = \text{Tr}(\rho) = 1$ . So  $\mu$  is a quantum measure that is not of the form  $\text{Tr}(\rho \cdot)$ . Dvurečenskij and Solovay proved that this is the only obstruction: a Hilbert space of dimension  $\lambda$  carries a quantum measure that is not of trace form if and only if some real-valued measurable cardinal is at most  $\lambda$  [Dvu93, Sol09]. Gleason's Theorem therefore holds verbatim for every Hilbert space whose dimension lies below the least real-valued measurable cardinal, in particular for every separable one. Theorem 1.2, on the other hand, holds in every dimension at least 3, with no set-theoretic hypothesis (see Remark 1.6 below).

If  $|I|$  is measurable, then  $\nu$  can be taken two-valued, that is, the indicator of a nonprincipal  $\sigma$ -complete ultrafilter  $\mathcal{U}$  on  $I$ , and then  $\mu(P) = \lim_{i, \mathcal{U}} \langle e_i | P | e_i \rangle$ . Even so, the measure  $\mu$  is not two-valued in general, because the numbers  $\langle e_i | P | e_i \rangle$  need not tend to 0 or 1 along  $\mathcal{U}$ . By Ulam's theorem [Ula30], a real-valued measurable cardinal is either at most  $2^{\aleph_0}$  or measurable. Solovay showed that adding  $\kappa$  random reals to a model with a measurable cardinal  $\kappa$  makes  $2^{\aleph_0} = \kappa$  real-valued measurable [Sol71], while the constructible universe has no real-valued measurable cardinals at all. So, relative to the consistency of a measurable cardinal, Gleason's Theorem for Hilbert spaces of dimension  $2^{\aleph_0}$  is independent of ZFC.


**Remark 1.6 (Ultrafilters on bases, and the Main Theorem in every dimension).**

In this remark we drop separability, and we weaken the notion of a two-valued measure as far as possible: let  $v: L(H) \rightarrow \{0, 1\}$  satisfy  $v(I) = 1$  and  $v(P \vee Q) = v(P) + v(Q)$  for orthogonal  $P, Q$ , with no countable additivity at all. Let  $\{e_i\}_{i \in I}$  be an orthonormal basis of  $H$ . The projections onto the closed spans of the sets  $\{e_j : j \in J\}$ , for  $J \subseteq I$ , form a Boolean algebra isomorphic to the power set of  $I$ , and  $v$  restricts to the indicator of an ultrafilter  $\mathcal{U}_v$  on  $I$ . If  $v$  is countably additive and  $I$  is countable, then  $\mathcal{U}_v$  is a  $\sigma$ -complete ultrafilter on a countable set, hence principal, and its generator is the unique winner of Lemma 2.1 below: the lemma is the principality of  $\sigma$ -complete ultrafilters on countable sets. On an uncountable basis, the ultrafilter  $\mathcal{U}_v$  may be nonprincipal; finite additivity permits any ultrafilter, and countable additivity permits any  $\sigma$ -complete one, which by Ulam's theorem is principal unless  $|I|$  is at least a measurable cardinal.

Nevertheless, Theorem 1.2 survives in every dimension at least 3, even for such finitely additive  $v$ , by amplifying a finite witness. In finite dimension, finite additivity is all of additivity, and the theorem applies as stated. If  $\dim H$  is infinite, write  $H = \mathbb{C}^3 \otimes K$  for a Hilbert space  $K$  of the same dimension. By Corollary 4.2 below, there are finitely many rays  $u_1, \dots, u_m$  of  $\mathbb{C}^3$  and orthonormal bases  $B_1, \dots, B_n$  drawn from them that admit no assignment taking the value 1 at exactly one ray of each basis. For each ray  $u$ , let  $W_u$  be the projection onto  $u \otimes K$ . If  $u \perp w$ , then  $W_u \perp W_w$ , and for each basis  $\{a, b, c\}$  among the  $B_k$  we have  $W_a + W_b + W_c = I$ . Finite additivity gives  $v(W_a) + v(W_b) + v(W_c) = 1$ , so  $u \mapsto v(W_u)$  is an assignment that the configuration forbids. So, in dimension at least 3, the lattice  $L(H)$  has no points in the strongest sense: it carries no two-valued measure, not even a finitely additive one, although each of its Boolean blocks carries as many ultrafilters as the power set of a basis does.

## 2. THE “ONLY ONE WINNER PER BASIS” LEMMA

The following elementary lemma plays a pivotal role in these notes.

**Lemma 2.1.** *Let  $v$  be a two-valued quantum measure on  $L(H)$  and  $\{e_i\}_{i \in I}$  an orthonormal basis of  $H$ . Exactly one index  $i \in I$  has  $v(P_{e_i}) = 1$ ; the rest have  $v(P_{e_i}) = 0$ .*

*Proof.* For  $i \neq j$  in  $I$ , orthogonality gives  $v(P_{e_i}) + v(P_{e_j}) = v(P_{e_i} \vee P_{e_j}) \in \{0, 1\}$ , so at most one index has value 1. Summing over the basis, we obtain  $\sum_i v(P_{e_i}) = v(I) = 1$ ; since each term is 0 or 1, exactly one equals 1. 


**Remarks 2.2.**

- (1) The proof of Lemma 2.1 does not use  $\dim H \geq 3$ ; the lemma holds for any orthonormal basis of cardinality at least 2, in any dimension. This dimension-independence might suggest that the lemma alone proves Theorem 1.2, which is not the case (see item 2 below).
- (2) The lemma only pins down, separately, one privileged ray in *each* orthonormal basis, and says nothing about how the privileged ray of one basis relates to that of a different, overlapping basis. To prove Theorem 1.2, we must make the choices in different bases interact, and these notes give two ways of doing so. Bell's argument (see Section 3 below) uses Gleason's Theorem to force the choices to fit together into one continuous function on the sphere  $S(H)$ , and then uses the connectedness of  $S(H)$  to break that

function against itself.<sup>3</sup> Kochen and Specker’s argument (see Section 5 below) uses instead a finite family of bases that overlap in a pattern that no choice of privileged rays can respect.

### 3. BELL’S 1964 PROOF OF THEOREM 1.2, BASED ON GLEASON’S THEOREM AND ELEMENTARY TOPOLOGY

In this section we present Bell’s proof of Theorem 1.2, which combines Gleason’s Theorem with the connectedness of the unit sphere. The Kochen–Specker proof, found independently in the same years, is given in Section 5; the history of both is traced in the Timeline (see Section 7 below).

*Proof of Theorem 1.2, via Gleason’s Theorem and connectedness.* Suppose that  $v$  is a two-valued quantum measure on  $L(H)$ . By Gleason’s Theorem, we have  $v(P) = \text{Tr}(\rho P)$  for some density operator  $\rho$ , so  $g(\psi) := v(P_\psi) = \langle \psi | \rho | \psi \rangle$  (see Observation 0.2) is continuous on  $S(H)$  (a matrix element of the fixed bounded operator  $\rho$ ) and  $\{0, 1\}$ -valued. Since  $S(H)$  is connected, a continuous  $\{0, 1\}$ -valued function on  $S(H)$  is constant; that is, there is  $c \in \{0, 1\}$  with  $g \equiv c$ . But by Lemma 2.1, on any orthonormal basis  $g$  equals 1 at exactly one point and 0 at the rest — so  $g$  takes *both* values on that basis, contradicting  $g \equiv c$ . ☹

💡 **Remark 3.1.** In the preceding proof, the hypothesis  $\dim H \geq 3$  is used only to invoke Gleason’s Theorem; nothing else depends on dimension.

💡 **Remark 3.2 (Sfumato).** Bell’s proof has a pictorial reading. Color each unit vector  $\psi$  by the value  $\mu(P_\psi)$  of a quantum measure  $\mu$ , from black (0) to white (1). A two-valued quantum measure would paint the sphere  $S(H)$  in pure black and white, and by Lemma 2.1 both colors appear on every orthonormal basis, so the picture would have hard contours. When  $\dim H \geq 3$ , Gleason’s Theorem allows only Leonardo’s *sfumato*: every quantum measure has the form  $\psi \mapsto \langle \psi | \rho | \psi \rangle$ , a continuous function on the path-connected space  $S(H)$ , so between a white vector and a black vector every shade of gray occurs, by the intermediate value theorem. For the pure state  $\rho = |u\rangle\langle u|$ , the gradation away from  $u$  is the Born rule  $\psi \mapsto |\langle u | \psi \rangle|^2$  itself. In dimension 2, where Gleason’s Theorem fails, hard contours are possible (see Remark 1.4).

🕒 **Historical note.** Gleason proved his theorem in 1957 to answer a question of Mackey’s about measures on closed subspaces [Mac57, Gle57], with no thought of hidden variables. In 1963 Jauch pointed out to Bell that the theorem bears on the hidden-variables question, and Bell turned this remark into the argument above; Bell thanks Jauch in a footnote of his paper [Bel66]. Bell wrote the paper in 1964, but the paper appeared only in 1966. The paper also criticizes two earlier impossibility arguments, von Neumann’s of 1932 and Jauch and Piron’s of 1963: each of them imposes a rule on the answers to questions that cannot be tested together, and nothing in physics justifies such a rule. The Timeline (see Section 7 below) tells the story in more detail.

🔗 **Provenance of the argument.** The proof of Theorem 1.2 given above is, in essential content, due to Bell. His own argument did not use the language of continuity or connectedness: he showed directly, using an auxiliary third orthogonal direction (available exactly because  $\dim H \geq 3$ ), that two rays with  $v$ -value 1 and 0 cannot be arbitrarily close; since both values must occur on every basis (see Lemma 2.1), two such close, oppositely valued rays are

<sup>3</sup>We observe that  $S(H)$  is connected; in fact, the sphere  $S(H)$  is even path-connected: if  $u, w \in S(H)$  and  $w \neq -u$ , the path  $t \mapsto \frac{(1-t)u+tw}{\|(1-t)u+tw\|}$  with  $t \in [0, 1]$  joins  $u$  to  $w$  inside  $S(H)$ , and the path  $\theta \mapsto e^{i\theta}u$  with  $\theta \in [0, \pi]$  joins  $u$  to  $-u$ .

unavoidable, a contradiction. What we have included here is the now-standard topological restatement of Bell's argument.

#### 4. OBTAINING A FINITE OBSTRUCTION VIA LOGICAL COMPACTNESS

Bell's proof of Theorem 1.2 in Section 3 proves nonexistence over the whole uncountable sphere without ever exhibiting a finite culprit. In this section we show that a finite culprit was *guaranteed* to exist the moment Bell's proof was known (see Corollary 4.2 below). The tool is compactness, which appears here in its logical and its topological form at once.

Throughout this section, we assume that  $\dim H = d$  with  $3 \leq d < \infty$ .

Let  $V = S(H)/\sim$  be the set of rays (unit vectors modulo phase factors, where  $u \sim e^{i\theta}u$  for  $\theta \in \mathbb{R}$ ), and let  $\mathcal{E}$  be the set of all orthonormal bases of  $H$ , each basis being a set of exactly  $d$  rays. Treat each ray  $u \in V$  as a propositional symbol  $\sigma_u$ , and to each basis  $e = \{u_1, \dots, u_d\} \in \mathcal{E}$  associate the propositional formula

$$\varphi_e := (\sigma_{u_1} \vee \dots \vee \sigma_{u_d}) \wedge \bigwedge_{1 \leq i < j \leq d} \neg(\sigma_{u_i} \wedge \sigma_{u_j}) \quad (\text{"exactly one } \sigma_{u_i} \text{ is true"}),$$

a formula in the  $d$  propositional symbols of  $e$  alone. The two-valued quantum measures on  $L(H)$  correspond exactly to the truth assignments to  $\{\sigma_u : u \in V\}$  that satisfy every  $\varphi_e$ . Indeed, by Lemma 2.1, a two-valued quantum measure  $v$  gives such an assignment, namely  $\sigma_u \mapsto v(P_u)$ . Conversely, let such an assignment be given. For  $P \in L(H)$ , let  $v(P)$  be the number of true  $\sigma_u$  with  $u$  in an orthonormal basis of the range of  $P$ . This number does not depend on the choice of basis: extending two bases of the range of  $P$  by one fixed basis of the range of  $P^\perp$  gives two bases of  $H$ , each with exactly one true ray. The same argument shows that  $v(P) \in \{0, 1\}$ , that  $v(I) = 1$ , and, using bases adapted to a family of pairwise orthogonal projections, that  $v$  is additive. So  $v$  is a two-valued quantum measure.

**The topological picture.** A truth assignment to the symbols  $\sigma_u$  is a point of the Cantor cube  $2^V = \{0, 1\}^V$ , which is compact by Tychonoff's theorem. For each basis  $e \in \mathcal{E}$ , the set

$$C_e = \{x \in 2^V : x \text{ satisfies } \varphi_e\}$$

depends only on the  $d$  coordinates indexed by the rays of  $e$ , so  $C_e$  is clopen. (In the language of Stone duality, the space  $2^V$  is the Stone space of the free Boolean algebra on the symbols  $\sigma_u$ , and  $C_e$  is the clopen set of the formula  $\varphi_e$ .) By the preceding paragraph, the two-valued quantum measures on  $L(H)$  correspond to the points of  $\bigcap_{e \in \mathcal{E}} C_e$ , and Theorem 1.2 says that this intersection is empty. Since  $2^V$  is compact, the finite intersection property then produces a finite subfamily of  $\{C_e : e \in \mathcal{E}\}$  with empty intersection. The compactness theorem of propositional logic, for the uncountable set of symbols  $\{\sigma_u : u \in V\}$ , is exactly this statement about  $2^V$  (see, for example, [End01]). The next proposition and corollary record the argument.

**Proposition 4.1.** *If every finite subset of bases  $\mathcal{E}_0 \subseteq \mathcal{E}$  admits some truth assignment to the finitely many rays occurring in  $\mathcal{E}_0$  satisfying  $\varphi_e$  for every  $e \in \mathcal{E}_0$ , then a two-valued quantum measure exists on all of  $L(H)$ .*

*Proof.* Let  $\mathcal{E}_0 \subseteq \mathcal{E}$  be finite. An assignment to the rays occurring in  $\mathcal{E}_0$  that satisfies every  $\varphi_e$  with  $e \in \mathcal{E}_0$  extends arbitrarily to a point of  $\bigcap_{e \in \mathcal{E}_0} C_e$ , so this intersection is nonempty. The sets  $C_e$  are closed in the compact space  $2^V$ , so the finite intersection property gives a point  $x \in \bigcap_{e \in \mathcal{E}} C_e$ . By the correspondence above, the point  $x$  is a two-valued quantum measure on  $L(H)$ . 

**Corollary 4.2.** *If  $3 \leq \dim H < \infty$ , then there are a finite set of rays and a finite set of orthonormal bases drawn from them that admit no assignment satisfying “exactly one per basis” on every one of those bases.*

*Proof.* By Theorem 1.2, proved in Section 3 via Gleason’s Theorem, no two-valued quantum measure exists on  $L(H)$ . By Proposition 4.1, some finite set of bases  $\mathcal{E}_0$ , drawn from finitely many rays, admits no assignment satisfying  $\varphi_e$  for every  $e \in \mathcal{E}_0$ . ☺



**Remark 4.3.** Corollary 4.2 says that a finite Kochen–Specker configuration exists, and the corollary derives this from Section 3 without constructing one. The derivation is non-constructive in two ways. Compactness of  $2^V$  for uncountable  $V$  is not provable in ZF; over ZF, this compactness is equivalent to the Boolean prime ideal theorem, a weak form of the Axiom of Choice. And the argument gives no bound on the size of the finite witness, nor any hint of where to find the witness.

The proof in the next section, in contrast, exhibits an explicit configuration of 18 vectors and uses no choice at all. Finding small explicit configurations took from 1967, when Kochen and Specker used 117 vectors, to 1996, when 18 vectors sufficed in dimension 4; in dimension 3 the smallest size is still unknown (see the [2024](#) entry of the Timeline in Section 7).

The appeal to choice is nevertheless harmless for the conclusion, though not for the proof. The statement of Corollary 4.2 asserts the existence of finitely many complex numbers satisfying finitely many polynomial equations, together with a finite combinatorial condition, so the statement is  $\Sigma_1^1$ . Since ZFC proves the statement and ZF proves that the constructible universe  $L$  satisfies ZFC, the statement holds in  $L$ , and  $\Sigma_1^1$  statements are upward absolute from  $L$  to the universe. Hence ZF proves Corollary 4.2. Choice is used by the compactness argument, but never by the theorem that the argument proves.

The topology also hides a trap. One might try to run the compactness argument over a countable dense set of rays, where compactness of  $2^\omega$ , which is provable in ZF, would suffice. In dimension 4 this happens to work, because the 18 vectors of Section 5 have coordinates in  $\{0, \pm 1\}$ . In dimension 3 it fails: Meyer [Mey99] showed that the rays through the rational points of the sphere  $S^2$  admit an assignment that takes the value 1 at exactly one ray of every orthonormal basis made of such rays. So in dimension 3 every finite witness involves irrational directions, and a dense set of rays need not carry any obstruction at all. Density is no substitute for continuity: Bell’s argument uses continuity on the whole sphere, and the assignment of a dense set need not extend continuously.

**The same picture, Boolean algebra by Boolean algebra.** The lattice  $L(H)$  is glued together from Boolean algebras: each family of commuting projections generates a Boolean subalgebra, and an orthonormal basis indexed by a set  $I$  generates one isomorphic to the power set of  $I$ , whose Stone space is  $\beta I$ . A finitely additive two-valued measure on  $L(H)$  (in finite dimension, simply a two-valued quantum measure) is exactly a family of ultrafilters, one on each Boolean subalgebra, that agree on common elements. In the product of the Stone spaces of all Boolean subalgebras, a compact space, each agreement condition defines a clopen set, and the two-valued measures are the points of the intersection of these clopen sets. For a single Boolean algebra, or for a chain of Boolean algebras, such an intersection is never empty. Theorem 1.2 says that for  $L(H)$  it is empty, and the finite intersection property says that finitely many agreement conditions already contradict one another, which is Corollary 4.2 once more. In infinite dimension, each Stone space  $\beta I$  has  $2^{2^{|I|}}$  points, and all of them together still cannot be made to agree (see Remark 1.6).

## 5. A FINITE PROOF OF THEOREM 1.2, AFTER KOCHEN AND SPECKER

In this section we give a proof of Theorem 1.2 that uses no measure theory and no continuity. A finite configuration, which can be checked by hand, replaces Bell's use of Gleason's Theorem and connectedness, and supplies the same missing ingredient: a way to force the separate "one winner per basis" choices of Lemma 2.1 to interact. We use the 18-vector configuration of Cabello, Estebaranz, and García-Alcaine [CEGA96], a descendant of Kochen and Specker's original 117-vector configuration (see the historical note at the end of this section). The configuration lives in  $\mathbb{R}^4$ , and at the end of the section we explain which dimensions the argument covers.

**Lemma 5.1 (Cabello–Estebaranz–García-Alcaine [CEGA96]).** *After normalization, the following 9 lists of 4 pairwise orthogonal vectors are orthonormal bases of  $\mathbb{R}^4$ . They are built from 18 distinct vectors in total, each of which occurs in exactly 2 of the 9 bases.*

$$\begin{aligned}
B_1 &= \{(0, 0, 0, 1), (0, 0, 1, 0), (1, 1, 0, 0), (1, -1, 0, 0)\} \\
B_2 &= \{(0, 0, 0, 1), (0, 1, 0, 0), (1, 0, 1, 0), (1, 0, -1, 0)\} \\
B_3 &= \{(1, -1, 1, -1), (1, -1, -1, 1), (1, 1, 0, 0), (0, 0, 1, 1)\} \\
B_4 &= \{(1, -1, 1, -1), (1, 1, 1, 1), (1, 0, -1, 0), (0, 1, 0, -1)\} \\
B_5 &= \{(0, 0, 1, 0), (0, 1, 0, 0), (1, 0, 0, 1), (1, 0, 0, -1)\} \\
B_6 &= \{(1, -1, -1, 1), (1, 1, 1, 1), (1, 0, 0, -1), (0, 1, -1, 0)\} \\
B_7 &= \{(1, 1, -1, 1), (1, 1, 1, -1), (1, -1, 0, 0), (0, 0, 1, 1)\} \\
B_8 &= \{(1, 1, -1, 1), (-1, 1, 1, 1), (1, 0, 1, 0), (0, 1, 0, -1)\} \\
B_9 &= \{(1, 1, 1, -1), (-1, 1, 1, 1), (1, 0, 0, 1), (0, 1, -1, 0)\}.
\end{aligned}$$

*Verification.* Orthogonality within each  $B_k$  is a check of  $\binom{4}{2} = 6$  dot products; for instance, for  $B_3$  we have  $(1, -1, 1, -1) \cdot (1, -1, -1, 1) = 1 + 1 - 1 - 1 = 0$ , and the other five products in  $B_3$  vanish in the same way. Table 1 lists, for each of the 18 vectors, the two bases that contain the vector. Since  $9 \text{ bases} \times 4 \text{ vectors} = 36 = 18 \times 2$ , the table accounts for every entry of every basis. 😊

| vector          | bases      | vector           | bases      | vector          | bases      |
|-----------------|------------|------------------|------------|-----------------|------------|
| $(0, 0, 0, 1)$  | $B_1, B_2$ | $(1, 0, -1, 0)$  | $B_2, B_4$ | $(1, 0, 0, 1)$  | $B_5, B_9$ |
| $(0, 0, 1, 0)$  | $B_1, B_5$ | $(1, -1, 1, -1)$ | $B_3, B_4$ | $(1, 0, 0, -1)$ | $B_5, B_6$ |
| $(1, 1, 0, 0)$  | $B_1, B_3$ | $(1, -1, -1, 1)$ | $B_3, B_6$ | $(0, 1, -1, 0)$ | $B_6, B_9$ |
| $(1, -1, 0, 0)$ | $B_1, B_7$ | $(0, 0, 1, 1)$   | $B_3, B_7$ | $(1, 1, -1, 1)$ | $B_7, B_8$ |
| $(0, 1, 0, 0)$  | $B_2, B_5$ | $(1, 1, 1, 1)$   | $B_4, B_6$ | $(1, 1, 1, -1)$ | $B_7, B_9$ |
| $(1, 0, 1, 0)$  | $B_2, B_8$ | $(0, 1, 0, -1)$  | $B_4, B_8$ | $(-1, 1, 1, 1)$ | $B_8, B_9$ |

TABLE 1. The 18 vectors of Lemma 5.1, each with the two bases that contain the vector.

Table 1 can be read as a graph: the vertices are the 9 bases, and each vector is an edge joining the two bases that contain the vector (see Figure 1). Each vertex has degree 4, because each basis has 4 vectors. A two-valued quantum measure would select, at every vertex, exactly one of the 4 edges, and an edge selected at one endpoint is selected at the other endpoint too, since the edge is a single ray. So the selected edges would form a perfect

matching of a graph with 9 vertices, which is impossible, because a perfect matching pairs off the vertices and 9 is odd. The proof below writes this parity argument as a double count.

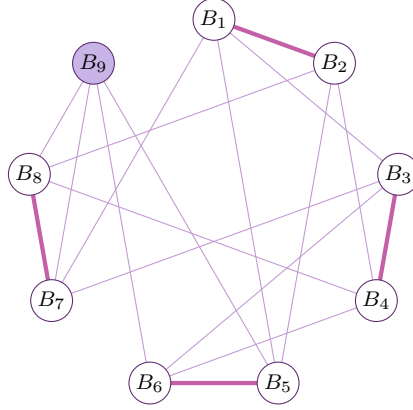


FIGURE 1. The 18 vectors as the edges of a 4-regular graph on the 9 bases. A two-valued quantum measure would be a perfect matching of this graph, and no graph with an odd number of vertices has one: a maximal matching (pink) always leaves a vertex uncovered (lavender).

*Proof of Theorem 1.2 for  $H = \mathbb{R}^4$ .* Suppose that  $v$  is a two-valued quantum measure on  $L(\mathbb{R}^4)$ . Restrict attention to the 18 rays of Lemma 5.1. Since each  $B_k$  is an orthonormal basis, additivity gives

$$\sum_{u \in B_k} v(u) = v(I) = 1 \quad \text{for each } k = 1, \dots, 9.$$

Summing over all 9 bases, we have

$$\sum_{k=1}^9 \sum_{u \in B_k} v(u) = 9.$$

We now reorganize the left side by vector instead of by basis. Each of the 18 vectors  $u$  occurs in exactly 2 of the bases  $B_k$ , so  $u$  contributes  $2v(u)$  to the double sum. Hence, we have

$$\sum_{k=1}^9 \sum_{u \in B_k} v(u) = \sum_u 2v(u) = 2 \sum_u v(u),$$

which is an even integer. This contradicts the fact that the double sum equals the odd integer 9. 😊

**Which dimensions the argument covers.** The same argument proves Theorem 1.2 for every real or complex Hilbert space  $H$  with  $\dim H \geq 4$ . Indeed, suppose that  $v$  is a two-valued quantum measure on  $L(H)$ . By Lemma 2.1, some unit vector  $e$  has  $v(P_e) = 1$ . Choose a 4-dimensional subspace  $K$  of  $H$  containing  $e$ , with projection  $P_K$ ; then  $v(P_K) = 1$ , because  $P_e \leq P_K$ . Fix an orthonormal basis  $f_1, \dots, f_4$  of  $K$ . The map  $(x_1, \dots, x_4) \mapsto \sum_i x_i f_i$  sends orthogonal vectors of  $\mathbb{R}^4$  to orthogonal vectors of  $K$ , so the 9 bases of Lemma 5.1 become 9 orthonormal bases of  $K$  with the same incidence pattern. For each of these bases, the 4 projections add up to  $P_K$ , so their values under  $v$  add up to  $v(P_K) = 1$ , and the double count above gives the same contradiction. The argument does not reach dimension 3, where

a configuration in  $\mathbb{R}^3$  is needed, such as Kochen and Specker’s 117 vectors or Peres’s 33 (see the [1991](#) entry of the Timeline). For dimension 3, these notes rely on Bell’s proof.

⌘ **Historical note.** Kochen and Specker’s 1967 proof [KS67] uses 117 vectors in  $\mathbb{R}^3$  (see Figure 2), and so the proof covers dimension 3 as well.

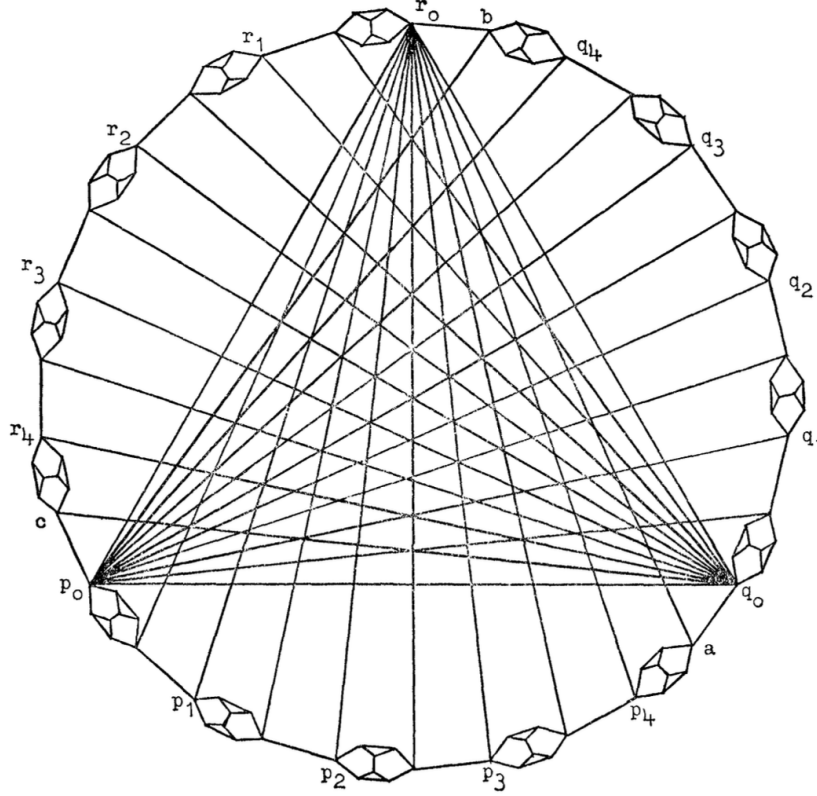


FIGURE 2. The graph  $\Gamma_2$  of Kochen and Specker’s 1967 paper [KS67, p. 69], whose 117 vertices are the vectors of their original proof; two vertices are joined by an edge when the corresponding vectors are orthogonal. The vertices are the points on the rim, after identifying  $p_0$  with  $a$ ,  $q_0$  with  $b$ , and  $r_0$  with  $c$ ; each of the fifteen small gadgets on the rim is a copy of their ten-vertex graph  $\Gamma_1$ .

Both authors were logicians. Ernst Specker (1920–2011), at ETH Zürich, had shown in 1953 that the Axiom of Choice fails in Quine’s New Foundations, and he had announced the present result without proof in 1960 [Spe60]. Simon Kochen (b. 1934) had written his Princeton thesis under Alonzo Church on ultraproducts, and was working with James Ax on the papers that became the Ax–Kochen theorem. Their formulation of the problem is accordingly logical. The questions that can be tested together form a Boolean algebra; these Boolean algebras, glued along their common parts, form a *partial Boolean algebra*; and the question is whether the whole structure admits a homomorphism to the two-element Boolean algebra. The 18-vector configuration used above was found in 1996 [CEGA96], after reductions by Peres and by Kernaghan (see the [1991](#), [1994](#), and [1996](#) entries of the Timeline).

## 6. QUANTUM MEASURES HAVE NO COUNTERPART OF DIRAC MEASURES AND DO NOT FORM A LATTICE

Classical probability has two features that are so familiar that they are easy to overlook. First, a single outcome decides every event: the events that contain a point  $\omega$  of the sample space form a principal ultrafilter, and the Dirac measure  $\delta_\omega$  is the indicator function of that ultrafilter. Second, any two finite measures on the same  $\sigma$ -algebra have a greatest lower bound and a least upper bound, so the measures form a lattice. In this section we show that quantum measures have neither feature when  $\dim H \geq 3$ . The failure of the first feature turns out to be a reformulation of the Kochen–Specker theorem. The failure of the second feature occurs already in dimension 2, and it has nothing to do with two-valued measures. Throughout this section, we assume that  $H$  is a Hilbert space with  $\dim H \geq 2$ .

**The classical properties (C1) and (C2).** We begin by stating the two classical features precisely. In classical probability, the events form a  $\sigma$ -algebra  $\Sigma$  of subsets of a sample space  $\Omega$ .

- (C1) Each outcome decides the Boolean operations on events. In terms of the Dirac measures, for every  $\omega \in \Omega$  and all  $A, B \in \Sigma$ , we have

$$\delta_\omega(A \cap B) = \min(\delta_\omega(A), \delta_\omega(B)), \quad \delta_\omega(\Omega \setminus A) = 1 - \delta_\omega(A). \quad (4)$$

(Each side of each equation equals 1 if  $\omega$  belongs to the event on the left-hand side, and 0 otherwise.)

- (C2) Let  $\mathcal{M}(\Sigma)$  be the set of finite positive measures on  $\Sigma$ , ordered by declaring  $\mu \leq \nu$  if  $\mu(A) \leq \nu(A)$  for every  $A \in \Sigma$ . Then  $(\mathcal{M}(\Sigma), \leq)$  is a lattice (see Proposition 6.1 below).

**The quantum questions (Q1) and (Q2).** To ask whether (C1) and (C2) survive in quantum mechanics, we replace the  $\sigma$ -algebra  $\Sigma$  by the lattice  $L(H)$ , and the probability measures by the quantum measures (see Definition 1.1). For a density operator  $\rho$ , we write  $\mu_\rho(P) = \text{Tr}(\rho P)$ . Each classical property then becomes a question.

- (Q1) For a quantum measure  $\mu$  on  $L(H)$ , the natural counterpart of the first equation in (4) is

$$\mu(P \wedge Q) = \min(\mu(P), \mu(Q)). \quad (5)$$

Is there a quantum measure  $\mu$  on  $L(H)$  that satisfies (5) whenever  $P, Q \in L(H)$  commute?

Two features of this formulation deserve a word of explanation. The second equation in (4) needs no counterpart, because every quantum measure satisfies  $\mu(P^\perp) = 1 - \mu(P)$  by additivity. The restriction to commuting pairs is the fair requirement: commuting projections lie in a common Boolean subalgebra of  $L(H)$ , and Boolean algebras are the setting of the classical equations (4). (If  $P$  and  $Q$  commute, the projections  $PQ$ ,  $P(I - Q)$ ,  $(I - P)Q$  and  $(I - P)(I - Q)$  are pairwise orthogonal with sum  $I$ ; the joins of subsets of these four projections form a Boolean subalgebra of  $L(H)$  containing  $P = PQ + P(I - Q)$  and  $Q = PQ + (I - P)Q$ .) Without this restriction, the question would be too easy: in Proposition 6.3 below, we show that no quantum measure satisfies (5) for all pairs  $P, Q \in L(H)$ .

- (Q2) For the counterpart of (C2), let  $\mathcal{C}(H)$  be the set of functions  $P \mapsto \text{Tr}(TP)$  on  $L(H)$ , where  $T$  ranges over the positive trace-class operators on  $H$ . (When  $\dim H \geq 3$ , Gleason’s Theorem shows that  $\mathcal{C}(H)$  is the set of nonnegative multiples of quantum measures.) As in (C2), we allow all nonnegative multiples of measures, not only the normalized ones, because a greatest lower bound of two probability measures is

in general not a probability measure; for example, the greatest lower bound of two distinct Dirac measures is 0. We order  $\mathcal{C}(H)$  pointwise:

$$\mu \leq \nu \quad \text{if } \mu(P) \leq \nu(P) \text{ for every } P \in L(H).$$

Is  $(\mathcal{C}(H), \leq)$  a lattice?

**The main result of this section.**

- When  $\dim H \geq 3$ , the answer to (Q1) is negative (see Theorem 6.7 below); when  $\dim H = 2$ , the answer is positive (see Remark 1.4).
- When  $\dim H \geq 2$ , the answer to (Q2) is negative (see Proposition 6.10 below).

Thus, when  $\dim H \geq 3$ , neither (C1) nor (C2) has a quantum counterpart.

**What the answers mean.** The negative answer to (Q1) is the Kochen–Specker theorem in disguise. In Lemma 6.2 below, we show that a quantum measure  $\mu$  satisfies (5) for all commuting  $P, Q$  if and only if  $\mu$  is two-valued, that is, if and only if  $\mu$  behaves like a Dirac measure. Hence, for  $\dim H \geq 3$ , the negative answer to (Q1) says exactly that there is no two-valued quantum measure on  $L(H)$ , which is Theorem 1.2. (In the terminology of Section 1, a two-valued quantum measure is a non-contextual hidden-variable assignment.) The measures used in physics fare even worse: when  $\dim H \geq 2$ , no measure  $\mu_\rho$  given by a density operator satisfies (5) for all commuting  $P, Q$  (see Corollary 6.8 below).

**Ultrafilters, principal and otherwise.** The answer to (Q1) is best read in the language of ultrafilters. On a Boolean algebra  $\mathcal{B}$ , a finitely additive two-valued measure  $v$  is the indicator function of the ultrafilter  $v^{-1}(1)$ , and every ultrafilter arises in this way, so the two-valued measures on  $\mathcal{B}$  are the points of the Stone space of  $\mathcal{B}$ . On the power set of a set  $\Omega$ , the Dirac measures are exactly the indicators of the principal ultrafilters. (Countable additivity is a strong demand: a countably additive two-valued measure on the power set of  $\Omega$  that vanishes on singletons is a nonprincipal  $\sigma$ -complete ultrafilter, and such ultrafilters exist only if  $|\Omega|$  is at least a measurable cardinal [Ula30]; compare Remark 1.5.)

In  $L(H)$ , a two-valued quantum measure  $v$  defines the set  $\mathcal{F}_v = \{P \in L(H) : v(P) = 1\}$ . This set is upward closed, contains exactly one of  $P$  and  $P^\perp$  for each  $P$ , and, by Lemma 6.2 below, is closed under meets of commuting pairs. These are the defining properties of an ultrafilter, read in the only Boolean algebras that  $L(H)$  contains, namely those generated by commuting families. So (Q1) asks whether  $L(H)$  has ultrafilters, and the Main Theorem says that, when  $\dim H \geq 3$ , the lattice  $L(H)$  has none: the quantum lattice has no points. The atoms of  $L(H)$  do generate principal filters,

$$\mathcal{F}_\psi = \{P \in L(H) : P \geq P_\psi\} = \{P \in L(H) : \psi \in \text{ran } P\} \quad (\psi \in S(H)),$$

which are closed under arbitrary meets. But  $\mathcal{F}_\psi$  contains  $P$  or  $P^\perp$  only when  $\psi$  is an eigenvector of  $P$ , so  $\mathcal{F}_\psi$  is not an ultrafilter. The failure of the principal filters to be ultrafilters is the superposition principle in lattice-theoretic dress. The pure state  $\mu_\psi(P) = \langle \psi | P | \psi \rangle$  is what remains of  $\mathcal{F}_\psi$ : we have  $\mu_\psi(P) = \|P\psi\|^2 = 1$  if and only if  $P\psi = \psi$ , that is, if and only if  $P \in \mathcal{F}_\psi$ , and on the questions that  $\mathcal{F}_\psi$  does not decide, the pure state  $\mu_\psi$  takes values strictly between 0 and 1.

The comparison also has a topological side. The Stone space of a Boolean algebra is compact and zero-dimensional, and its clopen sets are the events. The natural space of quantum points, the space of rays, is connected (see Proposition 6.4 below), so the space of rays has no nontrivial clopen sets. In Section 6.3 we turn this contrast into a proof.

The negative answer to (Q2) says that quantum measures are not ordered like classical measures. Recall that an *order isomorphism* between partially ordered sets  $(X, \leq)$  and  $(Y, \leq)$  is a bijection  $\Phi : X \rightarrow Y$  such that, for all  $x, x' \in X$ , we have  $x \leq x'$  if and only if  $\Phi(x) \leq \Phi(x')$ . For every  $\sigma$ -algebra  $\Sigma$  on a set  $\Omega$ , there is no order isomorphism between  $(\mathcal{C}(H), \leq)$  and  $(\mathcal{M}(\Sigma), \leq)$  (see Corollary 6.11 below). Since this holds even when  $\dim H = 2$ , where two-valued quantum measures exist (see Remark 1.4), the failure of (C2) is a phenomenon separate from the Kochen–Specker theorem.

**How the section is organized.** The proofs in this section do not use Theorem 1.2. We start on the classical side: in Section 6.1 we prove (C2); property (C1) needs no proof (see the parenthetical sentence after (4)). In Section 6.2 we prove Lemma 6.2, which turns (Q1) into a question about two-valued measures, and we show that the restriction to commuting pairs cannot be dropped. In Section 6.3 we answer (Q1) for  $\dim H \geq 3$ . The argument rests on a contrast between sliding and jumping: a unit vector can be moved continuously around the unit sphere, Gleason’s Theorem forces every quantum measure to change continuously along the way, and a two-valued measure can only jump between 0 and 1. In Section 6.4 we answer (Q2) with an explicit pair of commuting  $2 \times 2$  matrices. Finally, in Section 6.5 we use the compactness argument of Section 4 to show that finitely many projections in a 3-dimensional subspace of  $H$  already force the negative answer to (Q1).

**6.1. The classical case.** In this subsection we prove (C2). Given  $\mu, \nu \in \mathcal{M}(\Sigma)$ , the idea is to split  $\Omega$  into a part where  $\mu \geq \nu$  and a part where  $\mu \leq \nu$ , and to take the smaller of the two measures on each part. The Hahn decomposition theorem, which we recall below, provides the split, and in Proposition 6.1 below we show that the resulting measure is the greatest lower bound of  $\mu$  and  $\nu$ .

A *finite signed measure* on  $\Sigma$  is a countably additive function  $\lambda : \Sigma \rightarrow \mathbb{R}$ , for example  $\mu - \nu$ . Given a finite signed measure  $\lambda$ , a *Hahn set* for  $\lambda$  is a set  $S \in \Sigma$  such that  $\lambda(B) \geq 0$  for every measurable  $B \subseteq S$  and  $\lambda(B) \leq 0$  for every measurable  $B \subseteq \Omega \setminus S$ . The *Hahn decomposition theorem* says that every finite signed measure has a Hahn set (see, e.g., [Rud87, Theorem 6.14]).

Given  $\mu, \nu \in \mathcal{M}(\Sigma)$  and a Hahn set  $S$  for  $\mu - \nu$ , define measures  $\gamma, \eta \in \mathcal{M}(\Sigma)$  by

$$\gamma(A) = \nu(A \cap S) + \mu(A \setminus S), \quad \eta(A) = \mu(A \cap S) + \nu(A \setminus S) \quad (A \in \Sigma). \quad (6)$$

**Proposition 6.1.** *Let  $\mu, \nu \in \mathcal{M}(\Sigma)$  be given, and let  $S$  be a Hahn set for  $\mu - \nu$ . Then the measures  $\gamma$  and  $\eta$  defined in (6) are, respectively, the greatest lower bound and the least upper bound of  $\mu$  and  $\nu$  in  $(\mathcal{M}(\Sigma), \leq)$ .*

*Proof.* Since  $S$  is a Hahn set for  $\mu - \nu$ , we have  $\mu(B) \geq \nu(B)$  for every measurable  $B \subseteq S$  and  $\mu(B) \leq \nu(B)$  for every measurable  $B \subseteq \Omega \setminus S$ . Hence, for every  $A \in \Sigma$ , we have  $\nu(A \cap S) \leq \mu(A \cap S)$  and  $\mu(A \setminus S) \leq \nu(A \setminus S)$ , so  $\gamma(A) \leq \mu(A)$  and  $\gamma(A) \leq \nu(A)$ ; in other words, the measure  $\gamma$  is a lower bound of  $\mu$  and  $\nu$ . If  $\kappa \in \mathcal{M}(\Sigma)$  is another lower bound, then for every  $A \in \Sigma$  we have  $\kappa(A) = \kappa(A \cap S) + \kappa(A \setminus S) \leq \nu(A \cap S) + \mu(A \setminus S) = \gamma(A)$ . The proof that  $\eta$  is the least upper bound is the same, with the inequalities reversed. ☺

**6.2. Quantum measures that preserve commuting meets.** In this subsection we show that the condition in (Q1) is very restrictive: a quantum measure that satisfies the condition takes only the values 0 and 1, just as a Dirac measure does (see Lemma 6.2 below). Thus (Q1) asks exactly whether two-valued quantum measures exist. We also show that the restriction to commuting pairs in (Q1) cannot be dropped (see Proposition 6.3 below). To state these

results conveniently, we shall say that a quantum measure  $\mu$  on  $L(H)$  *preserves commuting meets* if  $\mu$  satisfies (5) whenever  $P, Q \in L(H)$  commute.

**Lemma 6.2.** *Let  $\mu$  be a quantum measure on  $L(H)$ . Then  $\mu$  preserves commuting meets if and only if  $\mu$  is two-valued.*

*Proof.* Additivity gives  $\mu(0) = 0$  and  $\mu(P^\perp) = 1 - \mu(P)$  for every  $P$ . Suppose that  $\mu$  preserves commuting meets. For every  $P$ , the projections  $P$  and  $P^\perp$  commute and  $P \wedge P^\perp = 0$ . Applying (5) to the pair  $P, P^\perp$  gives

$$\min(\mu(P), 1 - \mu(P)) = \min(\mu(P), \mu(P^\perp)) = \mu(P \wedge P^\perp) = \mu(0) = 0.$$

Since  $\min(r, 1 - r) > 0$  for every  $r \in (0, 1)$ , this forces  $\mu(P) \in \{0, 1\}$ .

Conversely, suppose that  $\mu$  is two-valued, and let  $P, Q \in L(H)$  be commuting projections. We first recall why  $PQ = P \wedge Q$ . Since  $P$  and  $Q$  commute, the operator  $PQ$  is self-adjoint and  $(PQ)^2 = PQ$ , so  $PQ$  is a projection. Its range lies in the ranges of both  $P$  and  $Q$  (because  $PQ = QP$ ), and  $PQ$  fixes every vector lying in both ranges. Hence  $PQ = P \wedge Q$ .

Next, we show that  $\mu(P \wedge Q) \leq \min(\mu(P), \mu(Q))$ . The projection  $P = PQ + P(I - Q)$  is a sum of two orthogonal projections, so additivity gives  $\mu(P) = \mu(P \wedge Q) + \mu(P(I - Q)) \geq \mu(P \wedge Q)$ . By symmetry, we also have  $\mu(Q) \geq \mu(P \wedge Q)$ .

Since  $\mu$  takes only the values 0 and 1, it remains to be shown that  $\mu(P \wedge Q) = 1$  whenever  $\mu(P) = \mu(Q) = 1$ . The projections  $P$  and  $(I - P)Q$  are orthogonal, so their sum is a projection and additivity gives  $\mu(P) + \mu((I - P)Q) \leq 1$ . Since  $Q = PQ + (I - P)Q$  is also a sum of two orthogonal projections, we have  $\mu((I - P)Q) = \mu(Q) - \mu(P \wedge Q)$ . Combining these two facts, we have

$$\mu(P) + \mu(Q) - \mu(P \wedge Q) \leq 1.$$

If  $\mu(P) = \mu(Q) = 1$ , this inequality gives  $\mu(P \wedge Q) \geq 1$ , and therefore  $\mu(P \wedge Q) = 1$ . ☺

Since the first equation in (4) holds for every pair of events, one might hope to drop the commutativity assumption in (Q1). The next proposition shows that, without this assumption, (Q1) has an immediate negative answer.

**Proposition 6.3.** *No quantum measure  $\mu$  on  $L(H)$  satisfies  $\mu(P \wedge Q) = \min(\mu(P), \mu(Q))$  for all  $P, Q \in L(H)$ .*

*Proof.* Suppose that  $\mu$  is such a quantum measure. In particular, the measure  $\mu$  preserves commuting meets, so  $\mu$  is two-valued by Lemma 6.2. Let  $\{e_i\}$  be an orthonormal basis of  $H$ . Additivity gives  $\sum_i \mu(P_{e_i}) = 1$ , so  $\mu(P_e) = 1$  for some  $e \in S(H)$ . Let  $K$  be a 2-dimensional subspace containing  $e$ , and let  $P_K$  be the projection onto  $K$ . Since  $P_K \geq P_e$ , we have  $\mu(P_K) = 1$ . For every orthonormal basis  $\{a, b\}$  of  $K$  we have  $P_a + P_b = P_K$ , so exactly one of  $\mu(P_a), \mu(P_b)$  equals 1. Fix an orthonormal basis  $\{a, b\}$  of  $K$  and let  $c = (a + b)/\sqrt{2}$  and  $d = (a - b)/\sqrt{2}$ , so that  $\{c, d\}$  is an orthonormal basis of  $K$  with no ray in common with  $\{a, b\}$ . Interchanging  $a$  and  $b$ , and  $c$  and  $d$ , if necessary, we may assume that  $\mu(P_a) = \mu(P_c) = 1$ . Then  $P_a \neq P_c$ . Two distinct rays meet in 0, so  $\mu(P_a \wedge P_c) = \mu(0) = 0$ , whereas  $\min(\mu(P_a), \mu(P_c)) = 1$ . ☹

**6.3. Topology and the negative answer to (Q1).** In this subsection we prove that the answer to (Q1) is negative when  $\dim H \geq 3$ . By Lemma 6.2, we must show that no two-valued quantum measure exists, and the argument plays continuity against jumps: Gleason's Theorem makes every quantum measure continuous on rays when  $\dim H \geq 3$ , whereas a two-valued quantum measure cannot be continuous on rays, because the unit sphere is connected and a two-valued quantum measure is not constant on rays.

We give  $L(H)$  the operator-norm topology, given by the metric  $d(P, Q) = \|P - Q\|$ . As in Section 0, for  $\psi \in S(H)$  we write  $P_\psi = |\psi\rangle\langle\psi|$  for the projection onto  $\mathbb{C}\psi$ , and we write  $\mu_\psi = \mu_{P_\psi}$  for the corresponding pure state, so that  $\mu_\psi(P) = \langle\psi|P|\psi\rangle$  (see Observation 0.2). The next proposition collects the three facts that we need: rank-one projections are close in norm exactly when they come from nearly parallel unit vectors, the rank-one projections form a connected space, and the number  $\mu_\rho(P)$  depends continuously on both the density operator  $\rho$  and the projection  $P$ .

**Proposition 6.4.**

- (1) For  $\psi, \varphi \in S(H)$  we have  $\|P_\psi - P_\varphi\| = (1 - |\langle\psi|\varphi\rangle|^2)^{1/2}$  and  $\|P_\psi - P_\varphi\|_1 = 2(1 - |\langle\psi|\varphi\rangle|^2)^{1/2}$ .
- (2) The set  $\{P_\psi : \psi \in S(H)\}$  of rank-one projections is a connected subset of  $L(H)$ .
- (3) For density operators  $\rho, \sigma$  and  $P, Q \in L(H)$  we have  $|\mu_\rho(P) - \mu_\sigma(Q)| \leq \|\rho - \sigma\|_1 + \|P - Q\|$ .

*Proof.* (1) The operator  $X = P_\psi - P_\varphi$  is self-adjoint, has trace 0, and has range in  $\text{span}(\psi, \varphi)$ . So the nonzero eigenvalues of  $X$  are  $\pm t$  for some  $t \geq 0$ , and, by Observation 0.2, we have  $2t^2 = \text{Tr } X^2 = \text{Tr } P_\psi + \text{Tr } P_\varphi - 2\text{Tr}(P_\psi P_\varphi) = 2 - 2|\langle\psi|\varphi\rangle|^2$ . Hence  $t = (1 - |\langle\psi|\varphi\rangle|^2)^{1/2}$ , and therefore  $\|X\| = t$  and  $\|X\|_1 = 2t$ .

(2) By (1), we have  $\|P_\psi - P_\varphi\| = (1 - |\langle\psi|\varphi\rangle|^2)^{1/2} \leq (2\|\psi - \varphi\|)^{1/2}$ , because  $1 - |\langle\psi|\varphi\rangle|^2 \leq 2(1 - \text{Re}\langle\psi|\varphi\rangle) = 2\text{Re}\langle\psi|\psi - \varphi\rangle \leq 2\|\psi - \varphi\|$ . So the map  $\psi \mapsto P_\psi$  is continuous, and the set of rank-one projections is the continuous image of the connected space  $S(H)$ .

(3) By the triangle inequality, we have  $|\text{Tr}(\rho P) - \text{Tr}(\sigma Q)| \leq |\text{Tr}((\rho - \sigma)P)| + |\text{Tr}(\sigma(P - Q))| \leq \|\rho - \sigma\|_1 \|P\| + \|\sigma\|_1 \|P - Q\|$ , and moreover  $\|P\| \leq 1$  and  $\|\sigma\|_1 = 1$ . ☺

**A first contrast with the classical case.** Before proving the main theorem of this subsection, we compare the pure states  $\mu_\psi$  with their classical counterparts, the Dirac measures. Let  $\Omega$  be a finite set with the discrete topology, and let  $\Sigma$  be the set of all subsets of  $\Omega$ . For every  $A \in \Sigma$ , the function  $\omega \mapsto \delta_\omega(A)$  takes only the values 0 and 1, and the set  $\{\omega : \delta_\omega(A) = 1\} = A$  is both open and closed in  $\Omega$ . The next proposition shows that both properties fail for the measures  $\mu_\psi$  on  $L(H)$ .

**Proposition 6.5.** *Let  $P \in L(H)$  be a projection with  $P \neq 0$  and  $P \neq I$ . The function  $\psi \mapsto \mu_\psi(P)$  on  $S(H)$  is continuous and takes every value in  $[0, 1]$ . The set  $\{\psi \in S(H) : \mu_\psi(P) = 1\}$  is closed, nonempty, and not open.*

*Proof.* Let  $f(\psi) = \mu_\psi(P) = \langle\psi|P|\psi\rangle$  for  $\psi \in S(H)$ . The function  $f$  is continuous, equal to 1 at the unit vectors in the range of  $P$  and to 0 at those in the range of  $P^\perp$ . By the intermediate value theorem along a path in  $S(H)$  joining two such vectors, exactly as in Remark 3.2, the function  $f$  takes every value in  $[0, 1]$ . The set  $f^{-1}(1)$  is closed by continuity, nonempty because  $P \neq 0$ , and proper because  $P \neq I$ . If  $f^{-1}(1)$  were also open, then  $S(H)$  would be disconnected. ☺

Proposition 6.5 fixes a projection  $P$  and lets the measure  $\mu_\psi$  vary. We now reverse the roles: we fix a measure and let the projection vary. The quantum measures that, like Dirac measures, take only the values 0 and 1 are the two-valued ones, and the next proposition shows that none of them varies continuously with the ray.

**Proposition 6.6.** *If  $v$  is a two-valued quantum measure on  $L(H)$ , then  $\psi \mapsto v(P_\psi)$  is not continuous on  $S(H)$ .*

*Proof.* Bell's proof of Theorem 1.2 in Section 3 uses Gleason's Theorem only to show that  $\psi \mapsto v(P_\psi)$  is continuous on  $S(H)$ . If the continuity of  $\psi \mapsto v(P_\psi)$  is assumed instead,

the rest of Bell's proof applies verbatim and yields a contradiction, using only Lemma 2.1 and the connectedness of  $S(H)$ ; in particular, this part of the argument does not require  $\dim H \geq 3$ . ☺

Combining Proposition 6.6 with Gleason's Theorem, which makes every quantum measure continuous on rays when  $\dim H \geq 3$ , we obtain the answer to (Q1).

**Theorem 6.7.** *A quantum measure on  $L(H)$  that preserves commuting meets exists if  $\dim H = 2$ , and does not exist if  $\dim H \geq 3$ .*

*Proof.* Suppose that  $\dim H \geq 3$ , and let  $\mu$  be a quantum measure that preserves commuting meets. By Lemma 6.2, the measure  $\mu$  is two-valued. By Gleason's Theorem, we have  $\mu = \mu_\rho$  for a density operator  $\rho$ , so  $\psi \mapsto \mu(P_\psi) = \langle \psi | \rho | \psi \rangle$  is continuous on  $S(H)$ . This contradicts Proposition 6.6.

If  $\dim H = 2$ , then, as in Remark 1.4, assign the value 1 to one ray in each pair of orthogonal rays and the value 0 to the other, and set  $v(0) = 0$  and  $v(I) = 1$ . This defines a two-valued quantum measure  $v$ , so  $v$  preserves commuting meets by Lemma 6.2. ☺

The second half of this proof shows what changes in dimension 2. There, Gleason's Theorem fails (see Remark 1.4), and the answer to (Q1) is positive; but by Lemma 6.2 and Proposition 6.6, every quantum measure that gives this positive answer is two-valued and therefore discontinuous on  $S(H)$ . The measures  $\mu_\rho$  used in physics (see Remark 0.1) are continuous, so in every dimension none of them satisfies the condition in (Q1).

**Corollary 6.8.** *For every density operator  $\rho$ , the quantum measure  $\mu_\rho$  does not preserve commuting meets.*

*Proof.* If  $\mu_\rho$  preserved commuting meets, Lemma 6.2 would make  $\mu_\rho$  two-valued. Since the function  $\psi \mapsto \mu_\rho(P_\psi) = \langle \psi | \rho | \psi \rangle$  is continuous, this would contradict Proposition 6.6. ☺

So far, the obstruction to (Q1) has come from the continuity of the measures. We end this subsection with an obstruction of a different kind, which comes from the meet operation itself. For a Dirac measure  $\delta_\omega$ , the first equation in (4) says that the number  $\delta_\omega(A \cap B)$  is computed from the numbers  $\delta_\omega(A)$  and  $\delta_\omega(B)$  by a fixed continuous function, namely  $\min$ . In Proposition 6.9 below, we show that no measure  $\mu_\rho$  given by a density operator has this property for any continuous function. The reason can be seen in a picture: the meet is not continuous for the norm topology. Two distinct lines through the origin meet only at 0, however small the angle between them, whereas a line meets itself in the whole line (see Figure 3).

**Proposition 6.9.** *Let  $P_n$  and  $P$  be rank-one projections with  $P_n \neq P$  and  $P_n \rightarrow P$  in  $L(H)$ . Then  $P_n \wedge P = 0$  and  $\|P_n \vee P - P\| = 1$  for every  $n$ , while  $P \wedge P = P \vee P = P$ . Moreover, for every density operator  $\rho$  there is no continuous  $m : [0, 1]^2 \rightarrow [0, 1]$  with  $\mu_\rho(Q \wedge R) = m(\mu_\rho(Q), \mu_\rho(R))$  for all  $Q, R \in L(H)$ .*

*Proof.* Since  $P_n \neq P$ , the ranges of  $P_n$  and  $P$  are distinct lines, so  $P_n \wedge P = 0$ . The join  $P_n \vee P$  is the projection onto a 2-dimensional space containing the range of  $P$ , so  $P_n \vee P - P$  is a rank-one projection and has norm 1.

Now let  $\rho$  be a density operator, and suppose that  $m : [0, 1]^2 \rightarrow [0, 1]$  is continuous and satisfies  $\mu_\rho(Q \wedge R) = m(\mu_\rho(Q), \mu_\rho(R))$  for all  $Q, R \in L(H)$ . Let  $\psi \in S(H)$  be given, choose a unit vector  $\varphi \perp \psi$ , and put  $Q_n = P_{\psi_n}$  with  $\psi_n = (\cos \frac{1}{n})\psi + (\sin \frac{1}{n})\varphi$ . Then  $Q_n \neq P_\psi$  and  $Q_n \rightarrow P_\psi$  by Proposition 6.4(1), so  $Q_n \wedge P_\psi = 0$  by the first part of this proof, and

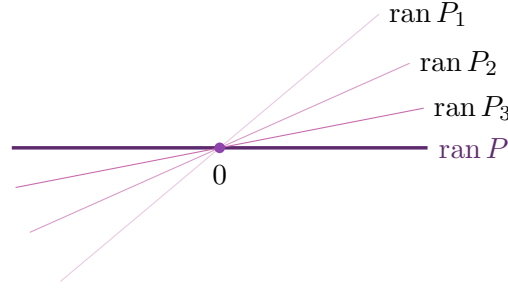


FIGURE 3. Rank-one projections  $P_n$  converging to a rank-one projection  $P$ , drawn through their ranges. For every  $n$ , the meet  $P_n \wedge P$  is 0 and the join  $P_n \vee P$  is the projection onto the plane spanned by the two lines; in the limit, both the meet and the join of  $P$  with itself are  $P$ .

$\mu_\rho(Q_n) \rightarrow \mu_\rho(P_\psi)$  by Proposition 6.4(3). Hence

$$0 = \mu_\rho(Q_n \wedge P_\psi) = m(\mu_\rho(Q_n), \mu_\rho(P_\psi)) \rightarrow m(\mu_\rho(P_\psi), \mu_\rho(P_\psi)) = \mu_\rho(P_\psi \wedge P_\psi) = \mu_\rho(P_\psi).$$

So  $\mu_\rho$  vanishes on every rank-one projection. Summing over an orthonormal basis gives  $\mu_\rho(I) = 0$ , a contradiction. ☹️

**6.4. The negative answer to (Q2).** In this subsection we prove that the answer to (Q2) is negative whenever  $\dim H \geq 2$  (see Proposition 6.10 below). The first step is to translate the order on  $\mathcal{C}(H)$  into the familiar order on self-adjoint operators, so that (Q2) becomes a question about positive matrices. Recall that  $\mathcal{C}(H) = \{\mu_T : T \geq 0 \text{ trace class}\}$ , where  $\mu_T(P) = \text{Tr}(TP)$ , and that  $\mathcal{C}(H)$  is ordered pointwise on  $L(H)$ . For positive trace-class operators  $S$  and  $T$ , we have  $\mu_S \leq \mu_T$  if and only if  $S \leq T$ . Indeed, if  $\mu_S \leq \mu_T$ , then  $\langle \psi | T - S | \psi \rangle = \mu_T(P_\psi) - \mu_S(P_\psi) \geq 0$  for every  $\psi \in S(H)$ , so  $T - S \geq 0$ ; and if  $T - S \geq 0$ , then  $\text{Tr}((T - S)P) = \text{Tr}(P(T - S)P) \geq 0$  for every  $P \in L(H)$ , so  $\mu_S \leq \mu_T$ .

To show that  $\mathcal{C}(H)$  is not a lattice, we need only exhibit two elements of  $\mathcal{C}(H)$  without a greatest lower bound. We choose two commuting operators, so that the failure cannot be attributed to noncommutativity of the operators. For orthonormal  $e_1, e_2 \in H$ , let

$$A = P_{e_1} + P_{e_2}, \quad D = 2P_{e_1} + \frac{1}{2}P_{e_2}. \quad (7)$$

**Proposition 6.10.** *Let  $e_1, e_2 \in H$  be orthonormal, and let  $A$  and  $D$  be the operators defined in (7). Then  $\mu_A$  and  $\mu_D$  have no greatest lower bound in  $\mathcal{C}(H)$ , although  $A$  and  $D$  commute.*

*Proof.* Let  $K = \text{span}(e_1, e_2)$ , and let  $\mu_T$  be a lower bound of  $\mu_A$  and  $\mu_D$  in  $\mathcal{C}(H)$ , so that  $0 \leq T \leq A$  and  $T \leq D$ . If  $\xi \perp K$ , then  $0 \leq \langle \xi | T | \xi \rangle \leq \langle \xi | A | \xi \rangle = 0$ , so  $T^{1/2}\xi = 0$  and hence  $T\xi = 0$ . Since  $T$  is self-adjoint, we have  $\langle \xi | Tx \rangle = \langle T\xi | x \rangle = 0$  for all  $x \in K$  and  $\xi \perp K$ . Hence  $T$  maps  $K$  into  $K$  and vanishes on  $K^\perp$ . We may therefore identify the lower bounds of  $\mu_A$  and  $\mu_D$  with positive  $2 \times 2$  matrices in the basis  $e_1, e_2$ . In this basis,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Throughout the argument we use the fact that a self-adjoint  $2 \times 2$  matrix is positive if and only if both diagonal entries and the determinant are nonnegative. In particular, the diagonal matrix

$$M = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

is a lower bound of  $A$  and  $D$ . Suppose that  $G$  were a greatest lower bound. Then  $M \leq G \leq A$  forces  $G_{11} = 1$ . The positive matrix  $G - M$  therefore has a zero  $(1,1)$  entry, and the nonnegativity of its determinant forces its off-diagonal entries to vanish. Then  $M \leq G \leq D$  forces  $G_{22} = \frac{1}{2}$ , so  $G = M$ . To reach a contradiction, consider

$$\tau = \begin{pmatrix} \frac{11}{12} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Then  $\tau \geq 0$  (since  $\det \tau = \frac{1}{6}$ ), the matrix  $A - \tau = \begin{pmatrix} 1/12 & -1/4 \\ -1/4 & 3/4 \end{pmatrix}$  is positive (since  $\det(A - \tau) = 0$ ), and the matrix  $D - \tau = \begin{pmatrix} 13/12 & -1/4 \\ -1/4 & 1/4 \end{pmatrix}$  is positive (since  $\det(D - \tau) = \frac{5}{24}$ ), so  $\tau$  is a lower bound of  $A$  and  $D$ . However, the matrix  $M - \tau = \begin{pmatrix} 1/12 & -1/4 \\ -1/4 & 1/4 \end{pmatrix}$  has determinant  $-\frac{1}{24} < 0$ , so  $\tau \not\leq M = G$ , which contradicts the choice of  $G$ . Hence no greatest lower bound exists. ☹

The proof of Proposition 6.10 shows where the argument of Proposition 6.1 breaks down. If we allowed only diagonal matrices, that is, operators that commute with  $P_{e_1}$  and  $P_{e_2}$ , then  $M$  would be the greatest lower bound of  $A$  and  $D$ , just as in the classical case (see Proposition 6.1). Indeed, we have  $\text{diag}(a, b) \leq A$  and  $\text{diag}(a, b) \leq D$  if and only if  $a \leq 1$  and  $b \leq \frac{1}{2}$ . The lower bound  $\tau$ , which shows that  $M$  is not the greatest lower bound, is not diagonal.

Proposition 6.10 has the following consequence (see the definition of order isomorphism at the beginning of this section).

**Corollary 6.11.** *Let  $\Sigma$  be a  $\sigma$ -algebra on a set  $\Omega$ . There is no order isomorphism between  $(\mathcal{C}(H), \leq)$  and  $(\mathcal{M}(\Sigma), \leq)$ .*

*Proof.* Suppose that  $\Phi : \mathcal{C}(H) \rightarrow \mathcal{M}(\Sigma)$  is an order isomorphism. Then  $\Phi^{-1}$  is also an order isomorphism. Let  $\mu, \nu \in \mathcal{C}(H)$  be given. By (C2) (see Proposition 6.1), the measures  $\Phi(\mu)$  and  $\Phi(\nu)$  have a greatest lower bound  $\gamma$  in  $\mathcal{M}(\Sigma)$ . We claim that  $\Phi^{-1}(\gamma)$  is a greatest lower bound of  $\mu$  and  $\nu$  in  $\mathcal{C}(H)$ . Since  $\gamma$  is a lower bound of  $\Phi(\mu)$  and  $\Phi(\nu)$ , the function  $\Phi^{-1}(\gamma)$  is a lower bound of  $\mu$  and  $\nu$ . If  $\kappa \in \mathcal{C}(H)$  is a lower bound of  $\mu$  and  $\nu$ , then  $\Phi(\kappa)$  is a lower bound of  $\Phi(\mu)$  and  $\Phi(\nu)$ , so  $\Phi(\kappa) \leq \gamma$ , and hence  $\kappa \leq \Phi^{-1}(\gamma)$ . This proves the claim. Thus any two elements of  $\mathcal{C}(H)$  have a greatest lower bound, which contradicts Proposition 6.10. ☹



**Remark 6.12 (Simplexes and the Bloch ball).** Two known theorems explain Proposition 6.10 geometrically; they are not used elsewhere in these notes. A compact convex subset of a finite-dimensional real vector space is a *simplex* if every point is a convex combination of extreme points in exactly one way, and a finite-dimensional real vector space ordered by a closed generating cone is a lattice if and only if a base of the cone is a simplex [Alf71]. Classically, the probability measures on a finite set  $\Omega$  form a simplex whose extreme points are the Dirac measures, that is, the principal ultrafilters once more. For  $H = \mathbb{C}^2$ , the base of  $\mathcal{C}(H)$  is the set of density operators. Every self-adjoint  $2 \times 2$  matrix of trace 1 can be written uniquely as  $\rho = \frac{1}{2}(I + x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3)$  with  $x \in \mathbb{R}^3$ , where

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix};$$

the eigenvalues of  $\rho$  are  $\frac{1}{2}(1 \pm |x|)$ , so  $\rho$  is a density operator if and only if  $|x| \leq 1$ . The density operators thus form a closed ball, the *Bloch ball*, whose extreme points, the pure states, fill the whole boundary sphere. The center  $\frac{1}{2}I$  is the midpoint of every diameter, since  $\frac{1}{2}I = \frac{1}{2}(P_\psi + P_{\psi'})$  for every orthonormal basis  $\{\psi, \psi'\}$  of  $\mathbb{C}^2$  (see Figure 4). So the Bloch ball is not a simplex, and the order is not a lattice. Kadison [Kad51] proved more: two self-adjoint

operators on a Hilbert space have a least upper bound in the self-adjoint operators only if the two operators are comparable.

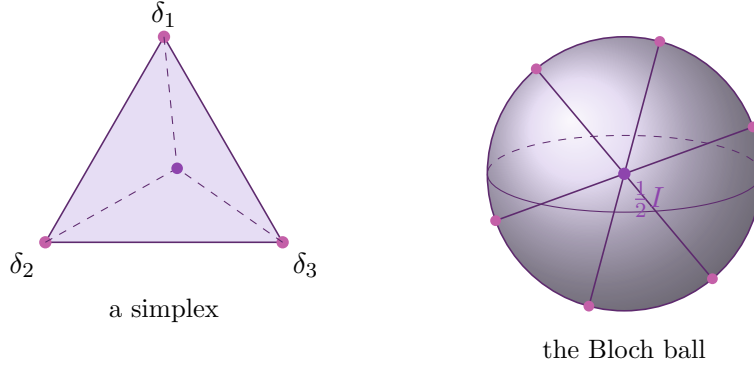


FIGURE 4. Left: the probability measures on a three-point space form a simplex, and each point is a unique convex combination of the Dirac measures at its vertices. Right: the density operators on  $\mathbb{C}^2$  form a ball whose extreme points (pink) are the pure states on the boundary sphere; the center  $\frac{1}{2}I$  is the midpoint of every diameter, so the decomposition into extreme points is far from unique.

**6.5. A finite witness for the negative answer to (Q1).** The proof of Theorem 6.7 applies Gleason's Theorem to all of  $L(H)$ , an infinite object. In this subsection we show that much less is needed: finitely many projections in a 3-dimensional subspace of  $H$  already force the negative answer to (Q1) (see Theorem 6.13 below). We derive the finite statement from Bell's argument and the compactness argument of Section 4, both applied in  $\mathbb{C}^3$ , and then transfer the finite statement to every Hilbert space of dimension at least 3.

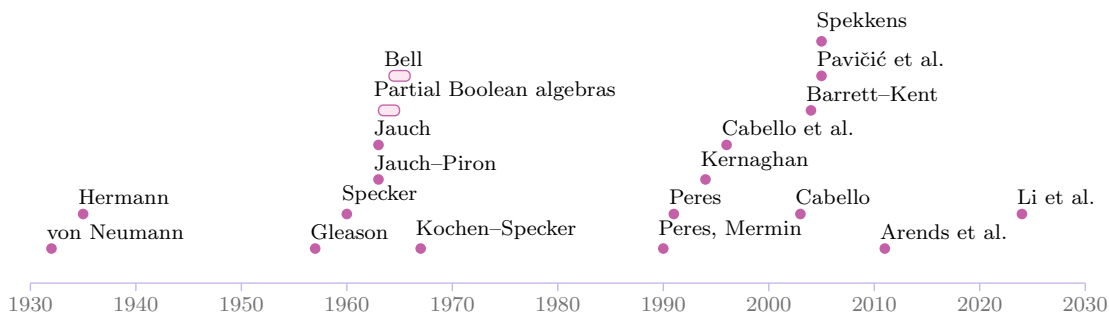
**Theorem 6.13.** *There is a finite family  $\mathcal{E}_0$  of orthonormal bases of  $\mathbb{C}^3$  such that no function from the rays occurring in  $\mathcal{E}_0$  to  $\{0, 1\}$  takes the value 1 at exactly one ray of each basis in  $\mathcal{E}_0$ . Consequently, if  $3 \leq \dim H \leq \infty$ , then no quantum measure on  $L(H)$  preserves commuting meets.*

*Proof. The finite family.* This is Corollary 4.2 with  $d = 3$ . Notice that the proof of Corollary 4.2 uses Theorem 1.2 only for  $H = \mathbb{C}^3$ , and only through Bell's argument in Section 3, which combines Gleason's Theorem in  $\mathbb{C}^3$  with the connectedness of the unit sphere; the only other ingredient is Proposition 4.1. A unitary map carries  $\mathcal{E}_0$  to a family with the same property, so  $\mathcal{E}_0$  can be placed in any 3-dimensional space.

*Transfer to  $H$ .* Let  $v$  be a quantum measure on  $L(H)$  that preserves commuting meets. By Lemma 6.2, the measure  $v$  is two-valued. Let  $\{e_i\}$  be an orthonormal basis of  $H$ . Additivity gives  $\sum_i v(P_{e_i}) = 1$ , so  $v(P_e) = 1$  for some  $e \in S(H)$ . Let  $K$  be a 3-dimensional subspace containing  $e$ , and let  $P_K$  be the projection onto  $K$ . Since  $P_e \leq P_K$ , we have  $v(P_K) = 1$ . By the first part of this proof, we may assume that  $\mathcal{E}_0$  consists of orthonormal bases of  $K$ . For each basis  $\{u_1, u_2, u_3\}$  of  $K$  in  $\mathcal{E}_0$  we have  $P_{u_1} + P_{u_2} + P_{u_3} = P_K$ , so  $v(P_{u_1}) + v(P_{u_2}) + v(P_{u_3}) = 1$ : exactly one of the three rays has value 1. Thus the restriction of  $v$  to the rays occurring in  $\mathcal{E}_0$  takes the value 1 at exactly one ray of each basis in  $\mathcal{E}_0$ , which is impossible by the choice of  $\mathcal{E}_0$ . This contradiction proves the theorem. ☺

The proof of Theorem 6.13 shows that the negative answer to (Q1) is witnessed by finitely many commuting families of three rank-one projections each, namely the images of the bases in  $\mathcal{E}_0$  under a unitary map of  $\mathbb{C}^3$  onto a single 3-dimensional subspace of  $H$ . Similarly, the witnesses in the proofs of Propositions 6.3 and 6.10 (the rays  $P_a, P_c$ , and the matrices  $M, \tau$ ) lie in a single 2-dimensional subspace of  $H$ .

## 7. TIMELINE



- 1932** • **von Neumann.** Von Neumann gives the first argument against hidden variables, in his book *Mathematische Grundlagen der Quantenmechanik* [vN32]. The argument assumes that the hidden value of a sum  $A + B$  of two observables is the sum of the hidden values of  $A$  and  $B$ , even when  $A$  and  $B$  cannot be measured together. Bell later put his objection to this assumption bluntly: “the proof of von Neumann is not merely false but foolish!” [MC88].
- 1935** • **Hermann.** Grete Hermann, a mathematician and philosopher who had written her doctoral thesis under Emmy Noether, objects to the same assumption in *Die naturphilosophischen Grundlagen der Quantenmechanik* [Her35]. Physicists largely ignored her objection until the 1970s [Jam74]; how closely her objection matches Bell’s is still debated [MS18].
- 1957** • **Gleason.** Gleason proves his theorem [Gle57] to answer a question of Mackey’s about measures on the closed subspaces of a Hilbert space [Mac57, Mac63]. The paper does not mention hidden variables.
- 1960** • **Specker.** In *Die Logik nicht gleichzeitig entscheidbarer Aussagen* (“The logic of propositions that cannot be decided simultaneously”) [Spe60], Specker announces that the closed subspaces of a three-dimensional Hilbert space cannot be mapped into a Boolean algebra in a way that respects the operations on orthogonal subspaces. He says only that “an elementary geometric argument” (*ein elementargeometrisches Argument*) shows this, and gives no proof.
- 1963** • **Jauch–Piron.** Jauch and Piron publish *Can Hidden Variables be Excluded in Quantum Mechanics?* [JP63], an argument against hidden variables in the lattice setting of these notes. The argument assumes that if two questions both have probability 1, then so does their meet. Bell’s 1966 paper objects that, for questions that cannot be tested together, the meet

cannot be tested by testing the two questions, so nothing in physics forces this assumption.

**1963** • **Jauch.** J. M. Jauch (1914–1974), director of the Institute of Theoretical Physics at the University of Geneva, points out Gleason’s Theorem to Bell. At the time, Jauch is supervising Piron’s doctoral thesis [Pir64], which describes the logic of quantum mechanics by the lattice of closed subspaces, so Gleason’s Theorem lies at the center of Jauch’s own research. Bell later recalled that Jauch’s point was this: Gleason’s Theorem needs additivity only for questions that can be tested together, which is exactly the weakening of von Neumann’s assumption that Bell wanted [Bel87]. Bell thanks Jauch in a footnote of [Bel66]: “I am much indebted to Professor Jauch for drawing my attention to this work.”

**1963–1965** • **Partial Boolean algebras.** Kochen and Specker present *Logical Structures Arising in Quantum Theory* at the 1963 Berkeley symposium on the theory of models, and *The Calculus of Partial Propositional Functions* at the 1964 Jerusalem congress; both appear in 1965 [KS65b, KS65a]. These papers introduce *partial Boolean algebras*: roughly, families of Boolean algebras, one for each set of questions that can be tested together, glued along their common parts. The application to hidden variables comes in their 1967 paper.

**1964–1966** • **Bell.** Bell, a particle physicist at CERN who worked on the foundations of quantum mechanics as a side interest, writes *On the Problem of Hidden Variables in Quantum Mechanics* [Bel66] in 1964, during a leave at the Stanford Linear Accelerator Center [Ber14]. The paper criticizes the arguments of von Neumann and of Jauch and Piron, and derives the nonexistence of non-contextual hidden variables from Gleason’s Theorem by an argument close to that of Section 3. Publication is delayed until 1966 (Rev. Mod. Phys. **38**, 447–452), so the paper appears between the two stages of Kochen and Specker’s work, and independently of both.

**1967** • **Kochen–Specker.** Kochen and Specker publish *The Problem of Hidden Variables in Quantum Mechanics* [KS67], a self-contained finite proof with 117 vectors in  $\mathbb{R}^3$ , written in the language of partial Boolean algebras (see Section 5).

**1990** • **Peres, Mermin.** Peres and, independently and shortly after, Mermin find the “magic square”: a  $3 \times 3$  array of observables of a pair of two-dimensional systems, in which the products along the rows and columns rule out every assignment of predetermined values  $\pm 1$ . The magic square proves a statement about observables rather than about rays, and, like the proof of Section 5, the magic square argument is a parity argument.

**1991** • **Peres.** Peres reduces Kochen and Specker’s construction to 33 vectors in  $\mathbb{R}^3$ , and gives a 24-vector construction in  $\mathbb{R}^4$  [Per91]. Conway and Kochen later find 31 vectors in  $\mathbb{R}^3$  (unpublished; reported in [Per93]).

- 1994** • **Kernaghan.** Kernaghan finds 20-vector configurations in  $\mathbb{R}^4$  inside Peres’s 24-vector configuration [Ker94].
- 1996** • **Cabello et al.** Cabello, Estebaranz, and García-Alcaine find the 18-vector configuration in  $\mathbb{R}^4$  used in Section 5 [CEGA96]. They started from the configurations of Peres and Kernaghan, and noted that none of their 18-vector configurations lies inside any of the 96 known 20-vector ones, “which probably explains why they were not obtained previously.”
- 2003** • **Cabello.** Cabello extends these ideas to measurements that are not given by projections, on a single two-dimensional system [Cab03] (see Appendix A below).
- 2004** • **Barrett–Kent.** Barrett and Kent analyze whether the theorem survives when measurements have only finite precision and answers may be random [BK04] (see Appendix A below).
- 2005** • **Pavičić et al.** Pavičić, Merlet, McKay, and Megill show, by an exhaustive computer search, that 18 is the smallest number of vectors for such a configuration in dimension 4 [PMMM05]. So the configuration of Section 5 is not only economical but optimal.
- 2005** • **Spekkens.** Spekkens proposes a notion of non-contextuality phrased in terms of experimental procedures rather than projections, which applies to imperfect measurements [Spe05] (see Appendix A below).
- 2011** • **Arends et al.** Arends, Ouaknine, and Wampler prove, by computer search, that every such configuration in dimension 3 has at least 18 vectors [AOW11].
- 2024** • **Li et al.** The smallest number of vectors in dimension 3 is still unknown: the best known configuration has 31 vectors (Conway and Kochen, reported in [Per93]), and a computer search has raised the lower bound to 24 [LBG24]. In the same year, Ramanathan studies assignments with values in  $[0, 1]$  instead of  $\{0, 1\}$  [Ram24] (see Appendix A below).

**7.1. Why the 1967 paper became the famous one.** By Bell’s proof in Section 3, the nonexistence of two-valued quantum measures follows in a few lines from Gleason’s Theorem of 1957. Yet every student of the foundations of physics learns the result under the names of Kochen and Specker. There are three reasons, and none of them is that Kochen and Specker found a deeper fact.

First, Gleason’s paper never mentions hidden variables. Gleason wrote the paper to answer a question of Mackey’s, and nothing in the paper singles out two-valued measures. For six years nobody noticed the connection (see the **1963** entry).

Second, Bell’s paper was delayed (see the **1964–1966** entry) and appeared only a year before Kochen and Specker’s.

Third, and most importantly, Kochen and Specker’s proof depends on nothing else. The proof is finite and can be checked by hand, so the reader need not trust Gleason’s Theorem, a substantial piece of analysis. Finiteness also had consequences of its own. The proof invited thirty years of shrinking the configuration, from 117 to 33 and then 31 vectors in  $\mathbb{R}^3$ , and from 24 to 20 to 18 in  $\mathbb{R}^4$ . A finite configuration also makes sense for structures that do not

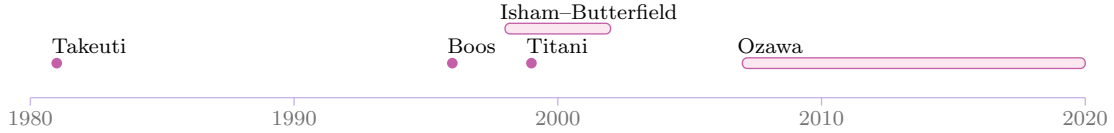
come from Hilbert spaces, where Gleason’s Theorem says nothing. Finally, a finite list of measurements can be carried out in a laboratory, whereas “all quantum measures on  $L(H)$  at once” cannot.

So Gleason had the result first, but Kochen and Specker gave the result a form that could be checked by hand, generalized, and tested in a laboratory.

## APPENDIX A. ATTEMPTS BEYOND CLASSICAL PROPOSITIONAL LOGIC

The proofs of Section 3 and Section 5 work throughout with propositional, two-valued ( $\{0, 1\}$ -valued) logic. Two natural ways to go beyond that setting are to keep the values binary but enrich the logic, adding bound variables and quantifiers, or to keep the propositional structure but enrich the values, allowing outcomes in  $[0, 1]$  rather than  $\{0, 1\}$ . This appendix records what has been done in each direction.

### A.1. Richer logics: quantifiers and first-order quantum logic.



**1981** • **Takeuti.** Gaisi Takeuti, already known for Boolean-valued analysis, publishes *Quantum Set Theory* [Tak81]. The construction mirrors the Scott–Solovay Boolean-valued models of ZFC exactly, with the complete Boolean algebra replaced by the ( $\sigma$ -complete) orthomodular lattice  $L$  of projections of a von Neumann algebra: a quantum universe  $V^{(L)}$  is built by transfinite recursion, and quantifiers receive a literal semantics, namely  $\llbracket (\forall x)\varphi(x) \rrbracket = \bigwedge_{x \in V^{(L)}} \llbracket \varphi(x) \rrbracket$ , an infimum taken in  $L$ . Real numbers in this universe become self-adjoint operators; this is genuine first-order (indeed set-theoretic) quantification, evaluated in a non-classical, non-Boolean truth-value algebra.

**1996** • **Boos.** William Boos publishes *Mathematical Quantum Theory I: Random Ultrafilters as Hidden Variables* [Boo96]. Where Takeuti changed the truth-value algebra, Boos brings to quantum theory the other main tool of modern set theory, Cohen’s method of forcing. His hidden variables are *random* ultrafilters, that is, generic objects in the sense of forcing with measure algebras, and he reads quantum probabilities through this lens. The paper is a rare attempt to address the hidden-variable question, and hence the setting of Theorem 1.2, with the methods of set theory rather than those of analysis.

**1998–2002** • **Isham–Butterfield.** Isham and Butterfield, joined for one paper by Hamilton, publish the four-part series *A Topos Perspective on the Kochen–Specker Theorem* [IB98, BI99, HIB00, IB02]. Motivated by the objection that the orthomodular lattice is too poorly behaved (non-distributive) to serve as a logic with an ordinary semantics, they replace it by the family of its Boolean subalgebras (“contexts”), each of which is classical. In their reformulation, Theorem 1.2 says that there is no family of points, one in the Stone space of each context, that agree whenever one context is contained in another:

this is the formulation of Section 4, and it is the statement of Section 6 that the quantum lattice has no points. Isham and Döring later extend this into a reformulation of physics, *A Topos Foundation for Theories of Physics* [DI08].

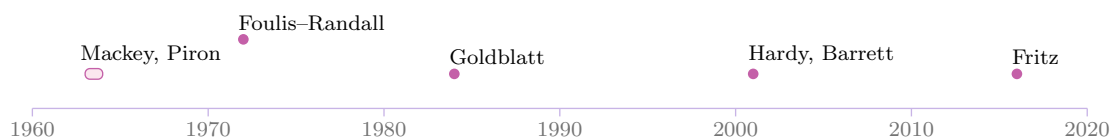
1999

**Titani.** Satoko Titani’s *A Lattice-Valued Set Theory* [Tit99] generalizes Takeuti’s construction from orthomodular lattices to arbitrary complete lattices as the truth-value algebra, placing quantum set theory inside a wider family alongside Heyting-valued (intuitionistic) set theories, with general completeness results.

2007–2020s

**Ozawa.** Masanao Ozawa turns quantum set theory from a suggestive construction into a working technical apparatus. *Transfer Principle in Quantum Set Theory* [Oza07] proves a rigorous dictionary carrying ZFC theorems about the reals over to theorems about self-adjoint operators inside  $V^{(L)}$ , in the spirit of the transfer principle of nonstandard analysis; *Orthomodular-Valued Models for Quantum Set Theory* [Oza09] tightens the model theory; later work reforms the semantics to recover De Morgan’s laws and extends the framework to quantum conditionals. Ozawa’s own motivation throughout is to state and prove a rigorous, quantifier-based version of Heisenberg’s uncertainty principle — exactly the kind of first-order reasoning a purely propositional logic cannot support.

A.1.1. *Classical first-order logic.* Classical first-order logic was tried in the most literal sense: treat the orthogonality relation on the closed subspaces of a Hilbert space as a first-order structure and ask the usual model-theoretic questions. The theorems of the field show that this approach cannot do the job.



1963–1964

**Mackey, Piron.** Before any of the above, Mackey’s axiomatic reconstruction of quantum mechanics [Mac63] and Piron’s representation theorem [Pir64] inaugurate a different and much more classical tradition: state the physical postulates (a set of “questions,” or an atomistic, orthomodular, irreducible lattice satisfying the covering law) as ordinary axioms in ordinary set-theoretic mathematics, with no non-classical logic anywhere, and *prove*, as a classical representation theorem, that any structure satisfying them is isomorphic to the subspace lattice of a generalized Hilbert space. The “quantum” content lives entirely in which structures satisfy the axioms, not in the logic used to reason about them.

1972 onward

**Foulis–Randall.** Foulis and Randall’s *manuals* and *test spaces* [FR72] continue this classical-metatheory tradition, starting from experiments rather than from lattices: outcomes and tests are ordinary sets, and non-Boolean

structure emerges only downstream, as a theorem about which propositions can be jointly tested, not as a feature of the logic used to derive that theorem.

- 1984

**Goldblatt.** Robert Goldblatt’s *Orthomodularity is not elementary* [Gol84] asks directly whether the class of Hilbert lattices is axiomatizable by classical first-order sentences in the language of the orthogonality relation, and proves that the answer is no: an infinite-dimensional pre-Hilbert space and the metric completion of that space are elementarily equivalent as first-order structures, yet only the complete one is orthomodular. Metric completeness — exactly the ingredient Theorem 1.2 and Gleason’s Theorem both quietly rely on (closedness of subspaces, convergence on  $S(H)$ ) — is provably invisible to classical first-order logic here.
  
- 2001, 2007

**Hardy, Barrett.** Hardy’s *Quantum Theory from Five Reasonable Axioms* [Har01] and Barrett’s *Information Processing in Generalized Probabilistic Theories* [Bar07] revive the program of Mackey and Piron in a modern form. The mathematics is again classical: quantum theory appears as one member of a family of probabilistic theories described by simple axioms about mixing states and combining systems.
  
- 2016

**Fritz.** Tobias Fritz’s *Quantum Logic is Undecidable* [Fri16], building on Slofstra’s work on embeddability problems for groups, shows that the classical first-order theory of complex Hilbert lattices is undecidable already at the level of quasi-identities: there is no algorithm deciding whether an implication between equations and orthogonality relations holds in every Hilbert space.



**Remark A.1.** So classical first-order logic was not overlooked. Tried on the most natural target (the orthogonality relation on subspaces), first-order logic was shown, by the theorems of the field itself, to be structurally too weak to pin down the class of Hilbert lattices at all (Goldblatt) and undecidable where applicable (Fritz). This double failure is a large part of why the two serious efforts at a genuinely quantified quantum logic (Takeuti’s line and the topos line) both abandoned classical semantics rather than pushing harder: Takeuti moved the non-classicality into the truth-value algebra, Isham–Butterfield moved the non-classicality into an intuitionistic internal logic, and both, not coincidentally, gained back exactly the property (extra structure that a bare classical first-order language cannot see) that Goldblatt’s theorem says is missing. The Mackey/Piron/Foulis–Randall/Hardy/Barrett line, meanwhile, never tried to axiomatize Hilbert lattices in first-order logic at all: that line uses classical first-order logic only as ordinary mathematics is always written, aimed at representation theorems about a differently chosen structure (a convex state space, a test space) rather than at characterizing subspace lattices up to elementary equivalence.

**A.2. Richer values:  $[0, 1]$ -valued and POVM-based assignments.** Theorem 1.2 rules out *sharp*, two-valued, non-contextual assignments. A natural physical question is whether an analogous no-go survives once outcomes, or the physical measurements themselves, are allowed to be *unsharp*. The answer depends on which of two things is fuzzed, and pulls in opposite directions.

Unsharp measurements are modeled as follows. An *effect* on  $H$  is a self-adjoint operator  $E$  with  $0 \leq E \leq I$ , that is, with spectrum in  $[0, 1]$ . A *positive-operator-valued measure* (POVM)

with finitely many outcomes is a finite family  $(E_1, \dots, E_n)$  of effects with

$$E_1 + \dots + E_n = I.$$

By the Born rule, a system in the state given by a unit vector  $\psi$  yields outcome  $i$  with probability  $\langle \psi | E_i | \psi \rangle$ ; the condition  $\sum_i E_i = I$  says exactly that these probabilities sum to 1. A projective measurement, i.e. an orthogonal decomposition of  $I$  into projections as in Definition 1.1, is the special case in which every  $E_i$  is a projection. Physically, a POVM is what one obtains by coupling the system to an auxiliary system and performing an ordinary projective measurement on the combined system (Naimark’s dilation theorem says every POVM arises this way). The essential new feature is that the effects of a POVM need not be mutually orthogonal, so a POVM may have more outcomes than  $\dim H$ , and a single effect may belong to many different POVMs.

*A.2.1. Fuzzing the measurements, keeping assignments sharp: the no-go strengthens, even at  $\dim H = 2$ .* The analogue of a two-valued quantum measure is now a map  $v$  from effects to  $\{0, 1\}$  such that, for every POVM  $(E_1, \dots, E_n)$ , exactly one  $E_i$  has  $v(E_i) = 1$ : each measurement has exactly one outcome. Non-contextuality is built into the requirement that  $v$  is a function of the effect alone, not of the POVM in which the effect appears. In  $\dim H = 2$  the projections give too few constraints for a contradiction: the only projective measurements are the pairs  $\{P, I - P\}$ , and a two-valued quantum measure exists (see Remark 1.4). Effects give many more constraints, because the same effect appears in many POVMs, with different companions. Cabello [Cab03] exhibited a finite family of POVMs on the qubit  $\mathbb{C}^2$  whose effects admit no such  $v$ . Thus the Kochen–Specker obstruction extends past sharp projections, into the regime  $\dim H = 2$ , which Theorem 1.2 does not reach.

*A.2.2. Fuzzing the assignments, keeping measurements sharp: the naive no-go can be evaded, and needs a better physical definition to survive.* Replace the deterministic  $v(P) \in \{0, 1\}$  by a probabilistic response function  $\xi_P(\lambda) \in [0, 1]$ : the probability of outcome 1 given the ontic state  $\lambda$ , a point of a measure space  $\Lambda$  that represents the complete physical state of the system, whether or not anyone knows  $\lambda$ . Barrett and Kent [BK04] observed that once a real experiment resolves outcomes only to finite precision, non-contextual hidden-variable models built from such fuzzy response functions can reproduce quantum statistics to arbitrary accuracy — so naive basis-independence becomes physically toothless once assignments are allowed to be fuzzy. Spekkens [Spe05] closed this gap with a stronger notion of non-contextuality: experimental procedures that no experiment can tell apart must be represented in the same way in the hidden-variable model. Under this notion, new impossibility theorems hold, some of them already in dimension 2.

*A.2.3. Extending the outcome alphabet directly.* Ramanathan [Ram24] proves a Kochen–Specker-type theorem for assignments valued in a finite alphabet  $\{0, p, 1 - p, 1\} \subseteq [0, 1]$  rather than only  $\{0, 1\}$ , for finite vector configurations in  $\mathbb{C}^3$ ; the case  $p = \frac{1}{2}$  rules out “fundamentally binary” hidden-variable theories.



**Remark A.2.** Fuzzing the *measurements* strengthens Theorem 1.2 (Cabello, reaching  $\dim H = 2$ ); fuzzing the *assignments* under the naive definition of non-contextuality weakens the theorem (Barrett–Kent), which is why Spekkens’s stronger definition was needed to recover a robust no-go there; extending the outcome set while keeping assignments deterministic recovers a close cousin of the original theorem outright (Ramanathan). In each case the substantive work is a fresh rigidity argument playing the role Gleason’s Theorem played earlier in these notes — never a restatement of the bare fact that local assignments respect the connectives.

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